Menstruation has a wide range of normal variation. The normal cycle length varies between 21 and 45 days. The length of blood loss varies between 3 and 7 days and the average blood loss per cycle is <80 ml – passage of blood clots or the use of more than six pads per day implies heavy bleeding, which needs evaluation. Rarely, it can indicate clotting disorders such as von Willebrand disease.

Summary

Puberty

- The first sign in females is breast development; in males it is testicular enlargement.
- In females the height spurt occurs shortly after breast development; in males it starts almost 18 months after the first signs of puberty.

Short stature

Short stature is usually defined as a height below the second centile (i.e. two standard deviations (SD) below the mean) or 0.4th centile (–2.6 SD). Only 1 in 50 children will be shorter than the 2nd centile and 1 in 250 (4 in 1000) shorter than the 0.4th centile. Most of these children will be normal, though short, with short parents, but the further the child is below these centiles, the more likely it is that there will be a pathological cause. However, the rate of growth (measured as height velocity; Fig. 11.1) may be abnormal long before a child's height falls below these values. This growth failure can be identified from the child's height falling across centile lines plotted on a height velocity chart (Fig. 11.1). This allows growth failure to be identified earlier, even though the child's height is still above the 2nd centile.

Measuring height velocity is a sensitive indicator of growth failure. Two *accurate* measurements at least 6 months but preferably a year apart allow calculation of height velocity in cm/year (Fig. 11.1). This is plotted at the midpoint in time on a height/velocity chart. A height velocity persistently below the 25th centile is abnormal and that child will eventually become short. A disadvantage of using height velocity calculations is that they are highly dependent on the accuracy of the height measurements and so tend not to be used outside specialist growth units.

The height centile of a child must be compared with the weight centile and an estimate of their genetic target centile and range calculated from the height of their parents. This is calculated as the mean of the father's and mother's height with 7 cm added for the mid-parental target height of a boy, and 7 cm subtracted for a girl. The 9th–91st centile range of this estimate is given by ±10 cm in a boy and ±8.5 cm in a girl (see examples in Fig. 11.9).

Most short children are psychologically well adjusted to their size. However, there may be problems from being teased or bullied at school, poor self-esteem and they are at a considerable disadvantage in most competitive sport. They are also assumed by adults to be younger than their true age and may be treated inappropriately.

Familial

Most short children have short parents and fall within the centile target range allowing for mid-parental height. Care needs to be taken, though, that both the child and a parent do not have a dominantly inherited growth disorder.

Intrauterine growth restriction (IUGR) and extreme prematurity

About one-third of children born with severe intrauterine growth restriction or who were extremely premature remain short. Growth hormone treatment may be indicated.

Constitutional delay of growth and puberty

These children have delayed puberty, which is often familial, usually having occurred in the parent of the same sex. It is commoner in males. It is a variation of the normal timing of puberty rather than an abnormal condition. It may also be induced by dieting or excessive physical training. An affected child will have delayed sexual changes compared with his peers, and bone age would show moderate delay. The legs will be long in comparison to the back. Eventually the target height will be reached. The condition may cause psychological upset. The onset of puberty can be induced with androgens or oestrogens.

Endocrine

Hypothyroidism, growth hormone (GH) deficiency, IGF-1 (insulin-like growth factor 1) deficiency and steroid excess are uncommon causes of short stature. They are associated with children being relatively overweight, i.e. their weight on a higher centile than their height.

Hypothyroidism

This is usually caused by autoimmune thyroiditis during childhood (see Ch. 25). This produces growth failure, usually with excess weight gain. It may go undiagnosed for many years and lead to short stature. When treated, catch-up growth rapidly occurs but often with a rapid entry into puberty that can limit final height. Congenital hypothyroidism is diagnosed soon after birth by screening and so does not result in any abnormality of growth.

Growth hormone deficiency

This may be an isolated defect or secondary to panhypopituitarism. Pituitary function may be abnormal in congenital mid-facial defects or as a result of a craniopharyngioma (a tumour affecting the pituitary region), a hypothalamic tumour or trauma such as head injury, meningitis and cranial irradiation. Craniopharyngioma usually presents in late childhood and may result in abnormal visual fields (characteristically a bitemporal hemianopia as it impinges on the optic chiasm), optic atrophy or papilloedema on fundoscopy. In growth

hormone deficiency, the bone age is markedly delayed. Laron syndrome is a condition due to defective growth hormone receptors resulting in growth hormone insensitivity. Patients with this condition have high growth hormone levels but low levels of the downstream active product of growth hormone known as insulin-like growth factor 1 (IGF-1) produced at the growth plate and in the liver. Rare abnormalities in the gene producing IGF-1 have also recently been discovered in children.

Corticosteroid excess, Cushing syndrome

This is usually iatrogenic, as corticosteroid therapy is a potent growth suppressor. This effect is greatly reduced by alternate day therapy, but some growth suppression may be seen even with relatively low doses of inhaled or topical steroids in susceptible individuals. Non-iatrogenic Cushing syndrome is very unusual in childhood and may be caused by pituitary or adrenal pathology. Growth failure may be very severe, usually with excess weight gain, although normalisation of body shape and height occurs on withdrawal of treatment or treatment of the underlying steroid excess. Cushing syndrome during puberty can result in permanent loss of height (see Ch. 25).

Nutritional/chronic illness

This is a relatively common cause of abnormal growth. These children are usually short and underweight: i.e. their weight is on the same or a lower centile than their height. Inadequate nutrition may be due to insufficient food, restricted diets or poor appetite associated with a chronic illness, or from the increased nutritional requirement from a raised metabolic rate. Chronic illnesses which may present with short stature include:

- Coeliac disease, which usually presents in the first 2 years of life, but can present late with growth failure. Coeliac disease may result in short stature without gastrointestinal symptoms
- Crohn disease
- Chronic renal failure – may be present in the absence of a history of renal disease.

Psychosocial deprivation

Children subjected to physical and emotional deprivation may be short and underweight and show delayed puberty. This condition may be extremely difficult to identify, but affected children show catch-up growth if placed in a nurturing environment.

Chromosomal disorder/syndromes

Many chromosomal disorders and syndromes are associated with short stature. Down syndrome is usually diagnosed at birth, but Turner (Fig. 8.6 and see Ch. 8), Noonan (Fig. 8.17) and Russell–Silver syndromes may present with short stature. Turner syndrome may be particularly difficult to diagnose clinically and *should be considered in all short females*.

Extreme short stature

There are a few rare conditions that cause extreme short stature in children. These include absolute resistance to growth hormone (Laron syndrome), and primordial dwarfism. Idiopathic short stature (ISS) refers to short stature that does not have Growth hormone resistance Primordial dwarfism Idiopathic short stature (ISS) a diagnostic explanation. In addition, abnormalities in a gene called *SHOX* (short stature homeobox) located on the X chromosome lead to severe short stature with skeletal abnormalities when present on both copies of the gene. Absence of one *SHOX* gene in Turner syndrome is thought to be the cause of short stature in this condition (and additional copies in Klinefelter syndrome produce taller than normal stature). Polymorphisms in this gene probably account for a proportion of idiopathic short stature.

Disproportionate short stature

This is confirmed by measuring:

- Sitting height – base of spine to top of head
- Subischial leg length – subtraction of sitting height from total height
- Limited radiographic skeletal survey to identify the skeletal abnormality.

Charts exist to assess the normality of body proportions. Conditions with abnormal body proportions are rare and may be caused by disorders of the formation of bone (skeletal dysplasias). They include achondroplasia and other short-limbed dysplasias. If the legs are extremely short, treatment by surgical leg lengthening may be appropriate. The back may be short from severe scoliosis or some storage disorders, such as the mucopolysaccharidoses.

Examination and investigation

Plotting present and previous heights and weights on appropriate growth charts, together with the clinical features, usually allows the cause to be identified without any investigations (Fig. 11.9a-i). Previous height and weight measurements should be available from the parent-held personal child health record. The bone age may be helpful, as it is markedly delayed in some endocrine disorders, e.g. hypothyroidism and growth hormone deficiency, and is used to estimate adult height potential. Investigations that may be indicated are shown in Table 11.1.

Growth hormone treatment of short stature

Growth hormone deficiency is treated with biosynthetic growth hormone, which is given by subcutaneous injection, usually daily. It is expensive and the management of growth hormone deficiency is undertaken at specialist centres. The best response is seen in children with the most severe hormone deficiency. Other indications include Turner syndrome, Prader–Willi syndrome (Fig. 8.19), chronic renal failure, *SHOX* (short stature homeobox) deficiency and intrauterine growth restriction (IUGR). In Prader–Willi syndrome (an imprinting disorder resulting in early hypotonia and feeding difficulties followed by short stature, obesity and learning difficulties), growth hormone improves muscular

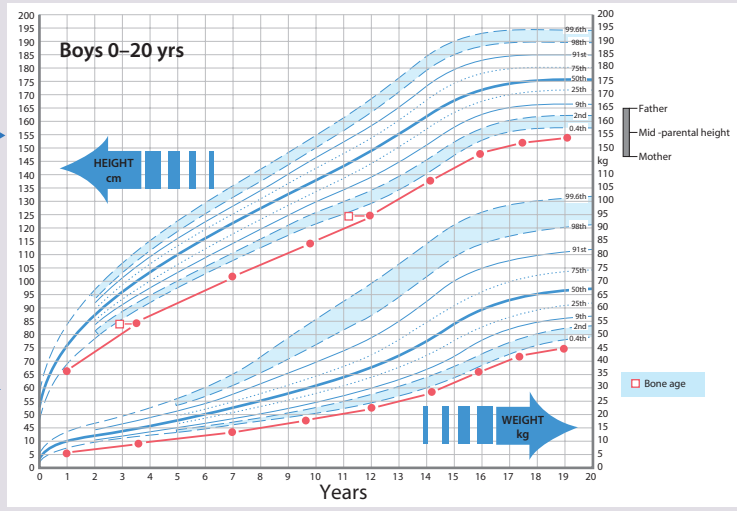
Causes & evaluation of short stature

(a)
Familial

Following growth centile within predicted range for parental height

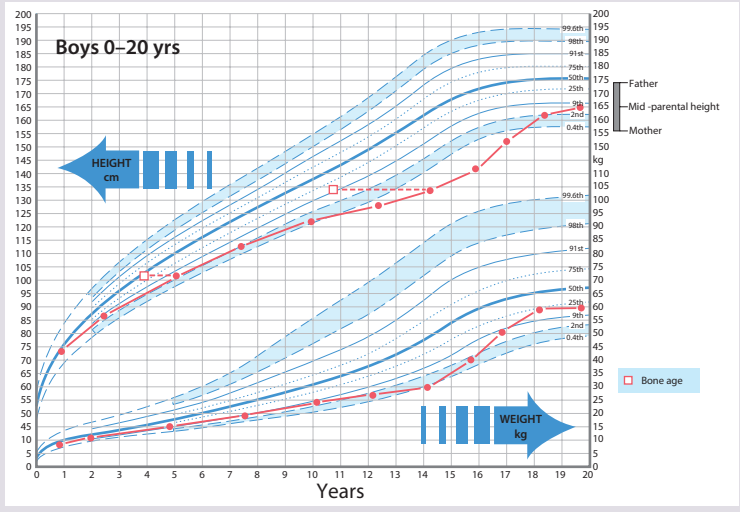
(b)
Severe intrauterine growth restriction or prematurity

Short from birth



(c)
Constitutional delay of growth and puberty

Short stature accentuated by delayed puberty. Delayed bone age



(d)
Endocrine
- growth hormone deficiency
- hypothyroidism
- steroid excess
  - iatrogenic
  - Cushing syndrome
  - IGF-1 deficiency

Falling off height centiles. Weight centile >height centile. Short and overweight Markedly delayed bone age.

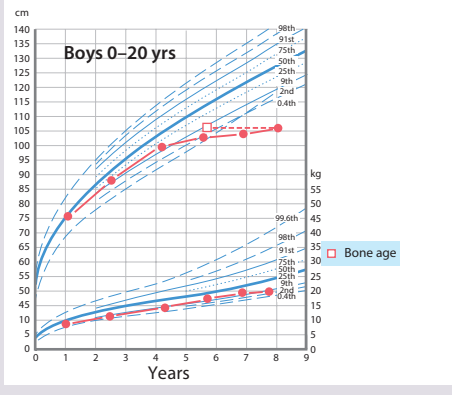


Figure 11.9 Causes of short stature.
Charts © Child Growth Foundation, reproduced with permission. Further supplies and information from www.healthforallchildren.co.uk.

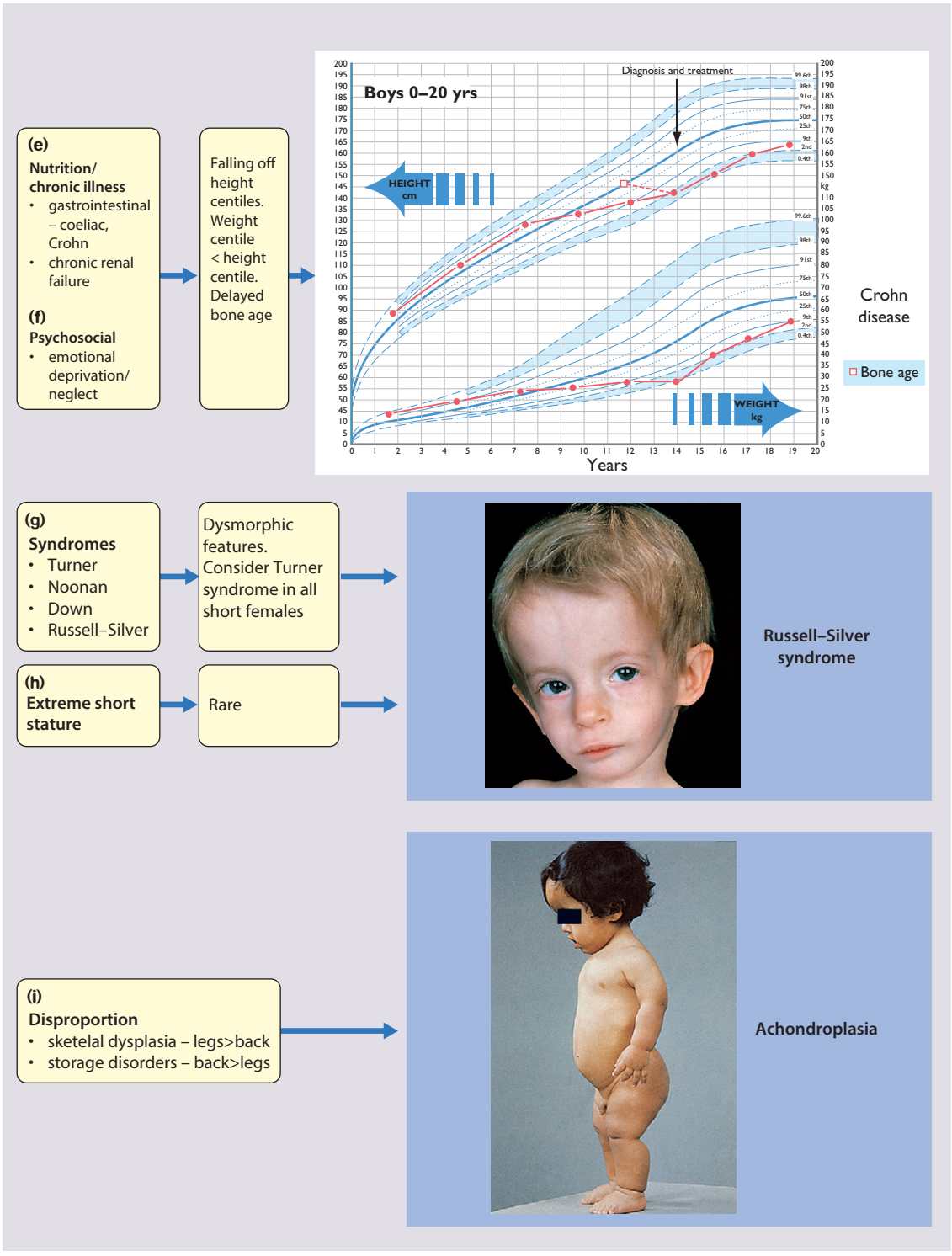


Figure 11.9, cont'd.

Table 11.1 Investigations considered for short stature

Investigation	Significance
X-ray of wrist and hand for bone age	Some delay in constitutional delay of growth and puberty Marked delay for hypothyroidism or growth hormone deficiency or other endocrine causes
Full blood count	Anaemia in coeliac or Crohn disease
Creatinine and electrolytes	Creatinine raised in chronic renal failure
Calcium, phosphate, alkaline phosphatise	Renal and bone disorders
Thyroid-stimulating hormone (TSH)	Raised in primary hypothyroidism
Karyotype	Turner syndrome shows 45XO, other chromosomal disorders
Endomysial and anti-tissue transglutaminase IgA antibodies	Usually present in coeliac disease
CRP (acute-phase reactant) and erythrocyte sedimentation rate (ESR)	Raised in Crohn disease
Growth hormone provocation tests (using insulin, glucagon, clonidine or arginine in specialist centres)	Growth hormone deficiency
IGF-1	Disorders of the growth hormone axis, including IGF-1 deficiency
0900 cortisol and dexamethasone suppression test	Cushing syndrome
MRI scan if neurological symptoms/signs	Craniopharyngioma or intracranial tumour
Limited skeletal survey	Skeletal dysplasia, scoliosis

strength and body composition as well as modestly improving final height. Recently, recombinant IGF-1 has been used to treat children with growth hormone resistance (e.g. Laron syndrome) and IGF-1 deficiency who would have previously not responded to growth hormone treatment. Recombinant IGF-1 therapy is still very expensive and is confined to a few specialised centres.

Tall stature

This is a less common presenting complaint than short stature, as many parents are proud that their child is tall. However, some adolescents (mainly females) become concerned about excessive height during their pubertal growth spurt. The causes are shown in Table 11.2. Most tall stature is inherited from tall parents. Obesity in childhood 'fuels' early growth and may result in tall stature; however, because puberty is often somewhat earlier than average, it does not increase final height.

Secondary endocrine causes are rare. Both congenital adrenal hyperplasia and precocious puberty lead to early epiphyseal fusion so that eventual height is reduced after an early excessive growth rate.

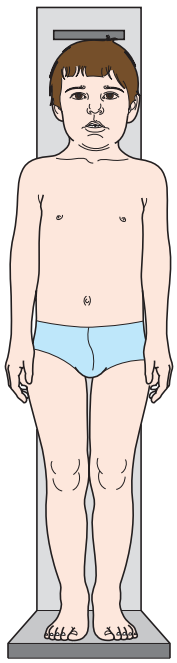
Marfan (a disorder of loose connective tissue) and Klinefelter (XXY – an excess of *SHOX* dose) syndromes

Table 11.2 Causes of excessive growth or tall stature

Familial	Most common cause
Obesity	Puberty is advanced, so final height centile is less than in childhood
Secondary	Hyperthyroidism
	Excess sex steroids – precocious puberty from whatever cause
	Excess adrenal androgen steroids – congenital adrenal hyperplasia
	True gigantism (excess GH secretion)
Syndromes	Long-legged tall stature:
	Marfan syndrome
	Homocystinuria
	Klinefelter syndrome (47 XXY and XXY karyotype)
	Proportionate tall stature at birth:
	Maternal diabetes
	Primary hyperinsulinism
	Beckwith syndrome
	Sotos syndrome – associated with large head, characteristic facial features and learning difficulties

Summary

Assessment of a child with short stature



Examination of the growth chart:
- Following growth centile lines for length/height, weight and head circumference?
  Consider familial, low birthweight, constitutional delay of growth and puberty, syndromes and skeletal dysplasias
- Growth failure with crossing of centile lines?
  Consider endocrine (including therapeutic corticosteroids), nutrition/chronic illness, psychosocial deprivation

Determine the mid-parental height
- For genetic target range

History
- Birth length, weight, head circumference and gestational age
- Pregnancy history: infection, intrauterine growth restriction, drug use, alcohol/smoking
- Feeding history
- Developmental milestones
- Family history of constitutional delay of growth and puberty or other diseases?
- Consanguinity pertaining to inherited conditions
- Features of chronic illness, endocrine causes, e.g. hypothyroidism, pituitary tumour, Cushing syndrome or psychosocial deprivation?
- Medications, e.g. corticosteroids?

Examination
- Dysmorphic features – chromosome/syndrome present? (But in Turner syndrome other stigmata may be absent)
- Chronic illness, e.g. Crohn, cystic fibrosis, coeliac disease?
- Evidence of endocrine causes?
- Disproportionate short stature from skeletal dysplasia?
- Pubertal stage?

Diagnosis
Cause can usually be determined from the above and no tests are required

both cause long-legged tall stature, and in XXY there is also infertility and learning difficulties.

Tall children may be disadvantaged by being treated as older than their chronological age. Excessive height in prepubertal or early pubertal adolescent females and males can be treated with oestrogen therapy and testosterone therapy, respectively, to induce premature fusion of the epiphyses, but as it produces variable results and has potentially serious side-effects, it is seldom undertaken. Surgical destruction of the epiphyses in the legs may also be considered in extreme cases.

Abnormal head growth

Most head growth occurs in the first 2 years of life and 80% of adult head size is achieved before the age of 5 years. This largely reflects brain growth, but small or large heads may be familial and the mid-parental head percentile may need to be calculated. At birth, the sutures and fontanelle are open. During the first few months of life, the head circumference may increase across centiles, especially if small for gestational age. The posterior fontanelle has closed by 8 weeks, and the anterior fontanelle by 12–18 months. If there is a rapid increase in head circumference, raised intracranial pressure should be excluded.

Microcephaly

Microcephaly, a head circumference below the 2nd centile, may be:

- Familial – when it is present from birth and development is often normal
- An autosomal recessive condition – when it is associated with developmental delay

Case History

11.1 Microcephaly

Figure 11.10 shows the head circumference chart of Tim, who was healthy and was developing normally. At 9 months of age, he was rushed to hospital as he was unrousable from profound hypoglycaemia secondary to the deliberate administration of insulin by his mother, who had diabetes. Although Tim was taken into care and had no further hypoglycaemic episodes, his head circumference shows cessation of growth. He has developed moderate learning difficulties and mild cerebral palsy.

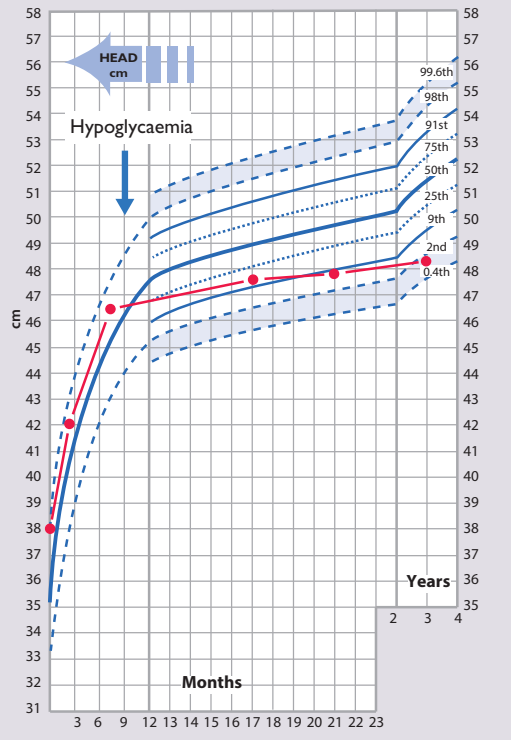


Figure 11.10 Tim's head circumference chart. Chart © Child Growth Foundation, reproduced with permission. Further supplies and information from www.healthforallchildren.co.uk.

- Caused by a congenital infection
- Acquired after an insult to the developing brain, e.g. perinatal hypoxia, hypoglycaemia or meningitis, when it is often accompanied by cerebral palsy and seizures (see Case History 11.1).

Macrocephaly

Macrocephaly is a head circumference above the 98th centile. The causes of a large head are listed in Box 11.1. Most are normal children and often the parents have large heads. A rapidly increasing head circumference, even if the head circumference is still below the 98th centile, suggests raised intracranial pressure and may be due to hydrocephalus, subdural haematoma or brain tumour. It must be investigated promptly by intracranial ultrasound if the anterior fontanelle is still open, otherwise by CT or MRI scan.

Box 11.1 Causes of a large head

- Tall stature
- Familial macrocephaly
- Raised intracranial pressure
- Hydrocephalus – progressive or arrested
- Chronic subdural haematoma
- Cerebral tumour
- Neurofibromatosis
- Cerebral gigantism (Sotos syndrome)
- CNS storage disorders, e.g. mucopolysaccharidosis (Hurler syndrome).

If an infant's head circumference is enlarging and crossing centile lines – check for raised intracranial pressure.

Asymmetric heads

Skull asymmetry may result from an imbalance of the growth rate at the coronal, sagittal or lambdoid sutures, although the head circumference increases normally. Occipital plagiocephaly, a parallelogram-shaped head with flattening of the back of the skull, is seen with increased frequency since the advice to parents that babies should sleep lying on their back to reduce the risk of sudden infant death syndrome. It improves with time as the infant becomes more mobile. Plagiocephaly is also seen in infants with hypotonia. Preterm infants may develop long, flat heads from lying on their sides for long periods on the hard surface of incubators unless provided with a soft surface to lie on and their head position is changed frequently (see Fig. 11.11). Under these circumstances, it is not associated with abnormal development.

Craniosynostosis

The sutures of the skull bones start to fuse during infancy but do not finally fuse until late childhood. Premature fusion of one or more sutures (craniosynostosis) may lead to distortion of the head shape. Craniosynostosis is usually localised (Box 11.2). It most often affects the sagittal suture, when it results in a long narrow skull (Fig. 11.12). Rarely it affects the lambdoid suture to result in skull asymmetry, which needs to be differentiated from plagiocephaly, where there is asymmetric flattening of one side of the skull from positional moulding.

Abnormal head shape

Box 11.2 Forms of craniosynostosis

Localised

- Sagittal suture – long narrow skull
- Coronal suture – asymmetrical skull
- Lambdoid suture – flattening of skull

Generalised

- Multiple sutures resulting in microcephaly and developmental delay
- Genetic syndromes, e.g. with syndactyly in Apert syndrome, with exophthalmos in Crouzon syndrome.

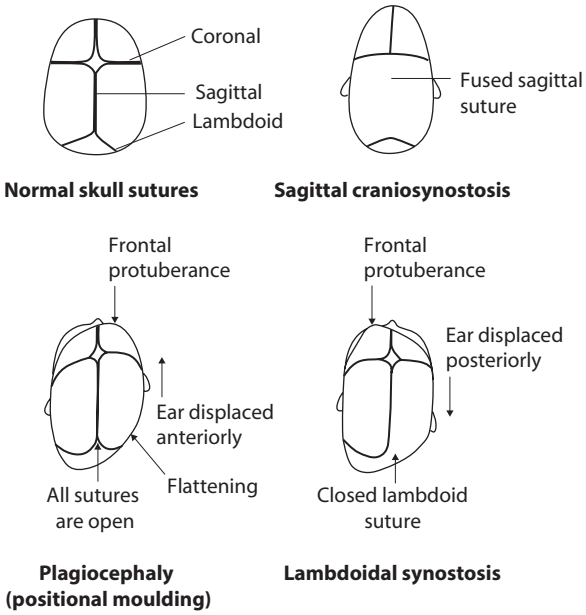


Figure 11.12 Differentiating craniosynostosis from plagiocephaly.



Figure 11.11 Long flat head of a preterm infant. This can be avoided by lying preterm infants on a soft surface and regularly changing their head position.



Figure 11.13 Crouzon syndrome showing the typical shallow orbits and exophthalmos. Craniofacial reconstructive surgery is required to prevent visual loss and cerebral damage from raised intracranial pressure and for cosmetic appearance.

Craniosynostosis may be generalised (Box 11.2), when it may be a feature of a syndrome (Fig. 11.13). The fused suture may be felt or seen as a palpable ridge and confirmed on skull X-ray or cranial CT scan. If necessary, the condition can be treated surgically because of raised intracranial pressure or for cosmetic reasons. Such operations are performed in specialist centres for craniofacial reconstructive surgery.

Premature sexual development

The development of secondary sexual characteristics before 8 years old in females and 9 years old in males is defined as outside the normal range in the UK. It may be due to:

- Precocious puberty when it is accompanied by a growth spurt
- Premature breast development (thelarche)
- Premature pubic hair development (pubarche).

Precocious puberty

Precocious puberty (PP) may be categorised according to the levels of the pituitary-derived gonadotropins, follicle-stimulating hormone (FSH) and luteinising hormone (LH), (Fig. 11.14) as:

- Gonadotropin-dependent (central, 'true' PP) from premature activation of the hypothalamic–pituitary–gonadal axis
- Gonadotropin-independent (pseudo, 'false' PP) from excess sex steroids.

Females

This is usually idiopathic or familial and follows the normal sequence of puberty. Organic causes are rare and are associated with:

- dissonance, when the sequence of pubertal changes is abnormal, e.g. isolated pubic hair with virilisation of the genitalia, suggesting excess

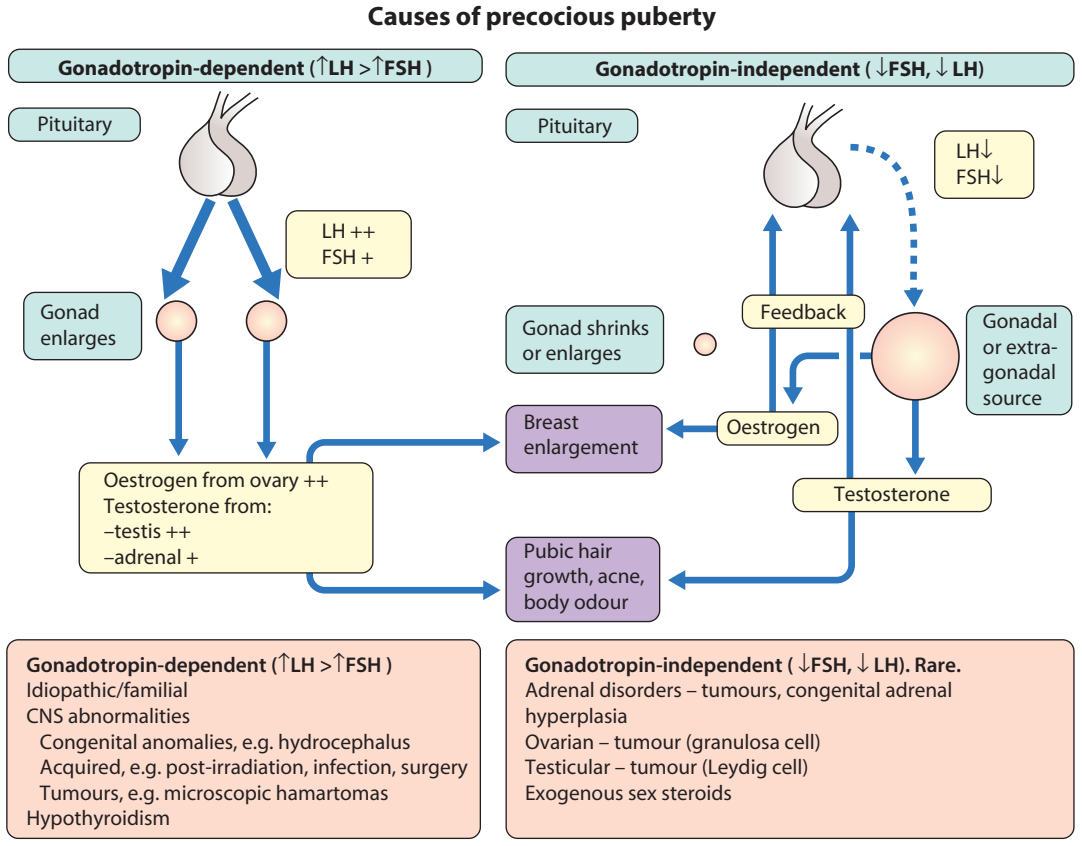


Figure 11.14 Causes of precocious puberty. (Courtesy of Dr Emma Rhodes.)

androgens from either congenital adrenal hyperplasia or an androgen-secreting tumour
- rapid onset
- neurological symptoms and signs, e.g. neurofibromatosis.

Ultrasound examination of the ovaries and uterus is helpful in establishing the cause of precocious puberty. In the premature onset of normal puberty, multicystic ovaries and an enlarging uterus will be identified.

Precocious puberty in females is commonly due to the premature onset of normal puberty.

Males (see Case History 11.2)

This is uncommon and usually has an organic cause, particularly intracranial tumours. Examination of the testes may be helpful:

- Bilateral enlargement suggests gonadotropin release, usually from an intracranial lesion
- Small testes suggest an adrenal cause (e.g. a tumour or adrenal hyperplasia)
- A unilateral enlarged testis suggests a gonadal tumour.

Tumours in the hypothalamic region are best investigated by cranial MRI scan.

Central precocious puberty in males more often has an organic cause.

Management

The management of precocious puberty is directed towards:

- detection and treatment of any underlying pathology, e.g. intracranial tumour in males, and reducing the rate of skeletal maturation if necessary. Skeletal maturation is assessed by bone age. An early growth spurt may result in early cessation of growth and a reduction in adult height.
- addressing psychological/behavioural difficulties associated with early progression through puberty.

Deciding whether to treat a girl who is simply going through puberty early needs further consideration. If treatment is required for gonadotropin-dependent disease, gonadotropin-releasing hormone (GnRH) analogues are the treatment of choice. In gonadotropin-independent cases, the source of excess sex steroids needs to be identified. Inhibitors of androgen or oestrogen production or action (e.g. medroxyprogesterone acetate, cyproterone acetate, testolactone, ketoconazole) may be used.

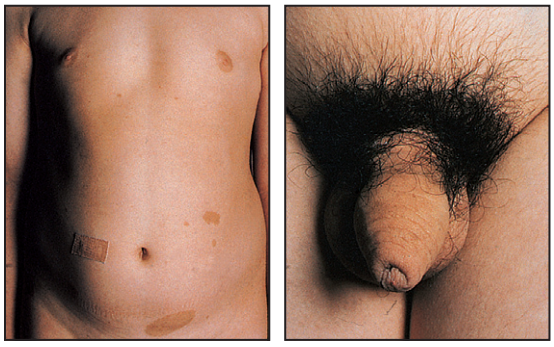
Premature breast development (thelarche)

This usually affects females between 6 months and 2 years of age. The breast enlargement may be asymmetrical and rarely progresses beyond stage 3. It is

Case History

11.2 Precocious puberty in a boy

This 6-year-old boy presented with precocious puberty (Fig. 11.15a,b). He was noted to have multiple café-au-lait spots consistent with a diagnosis of neurofibromatosis type 1. An MRI scan showed a mass in the hypothalamus which proved to be an optic glioma. He was treated with radiotherapy, although full remission was not possible to achieve. The site of injection of gonadotropin super-agonist treatment to suppress his sexual development is covered by the plaster.



(a) (b)

Figure 11.15 (a) Multiple café-au-lait spots. Neurofibromatosis type 1 was diagnosed. **(b)** Genitalia showing stage 3 genitalia and pubic hair with 12 ml testicles bilaterally. He also had adult body odour. (From Wales JKH, Rogol AD, Wit JM. 2003. *Pediatric Endocrinology and Growth*. Saunders, London, with permission.)

differentiated from precocious puberty by the absence of axillary and pubic hair and of a growth spurt. It is non-progressive and self-limiting. Investigations are not usually required (see Case History 11.3).

Premature pubarche (adrenarche)

This occurs when pubic hair develops before 8 years of age in females and before 9 years in males but with no other signs of sexual development. It is most commonly caused by an accentuation of the normal maturation of androgen production by the adrenal gland (adrenarche). It is more common in Asian and Afro-Caribbean children. There may be a slight increase in growth rate. It is usually self-limiting. An ultrasound scan of the ovaries and uterus and a bone age should be obtained to exclude central precocious puberty. A more aggressive course of virilisation would suggest late-onset non-salt-losing congenital adrenal hyperplasia (CAH) or an adrenal tumour. Obtaining a urinary steroid profile helps differentiate premature pubarche from late onset CAH or an adrenal tumour. Children who develop premature pubarche are at an increased risk of developing polycystic ovarian syndrome (PCOS) in later life.

Delayed puberty

Delayed puberty is often defined as the absence of pubertal development by 14 years of age in females and 15 years in males. The causes of delayed puberty are listed in Box 11.3.

In contrast to precocious puberty, the problem is more common in males, in whom it is mostly due to constitutional delay in growth and puberty (CDGP). This is often familial, usually having occurred in the parent of the same sex. It may also be induced by

Case History

11.3 Premature thelarche

This 18-month-old female developed enlargement of both breasts (Fig. 11.16). There was no pubic hair growth, sweatiness or body odour and her height was in the mid-parental range. Her bone age was only mildly advanced (21 months) and a pelvic ultrasound showed a prepubertal uterus, small volume ovaries with two cysts in the left ovary. Her subsequent growth rate was normal. A diagnosis of premature thelarche was made.



Figure 11.16 Premature breast development in an 18-month-old girl. The absence of a growth spurt and axillary and pubic hair differentiates it from precocious puberty. It is self-limiting and often resolves. (From Wales JKH, Rogol AD, Wit JM. 2003. *Pediatric Endocrinology and Growth*. Saunders, London, with permission.)

Box 11.3 Causes of delayed puberty

Constitutional delay of growth and puberty/familial

By far the commonest

Low gonadotropin secretion (hypogonadotropic hypogonadism)

- Systemic disease
  - Cystic fibrosis, severe asthma, Crohn disease, organ failure, anorexia nervosa, starvation, excess physical training
- Hypothalamo-pituitary disorders
  - Panhypopituitarism
  - Isolated gonadotropin or growth hormone deficiency
  - Intracranial tumours (including craniopharyngioma)
  - Kallmann syndrome (LHRH deficiency and inability to smell)
- Acquired hypothyroidism

High gonadotropin secretion (hypergonadotropic hypogonadism)

- Chromosomal abnormalities
  - Klinefelter syndrome (47 XXY)
  - Turner syndrome (45 XO)
- Steroid hormone enzyme deficiencies
- Acquired gonadal damage
  - Post-surgery, chemotherapy, radiotherapy, trauma, torsion of the testis, autoimmune disorder.

dieting or excessive physical training. Children affected with constitutional delay in growth and puberty are short during childhood, with a delay in sexual maturation and have delayed skeletal maturity on bone age. The legs will be long in comparison to the back (eunuchoid body habitus). Eventually the target height will be reached as growth in affected children will continue for longer than in their peers. The condition may cause considerable psychological upset from teasing, poor self-esteem and disadvantage in competitive sport.

In boys, assessment includes:

- Pubertal staging, especially testicular volume
- Identification of chronic systemic disorders.

In girls, karyotype should be performed to identify Turner syndrome, and thyroid and sex steroid hormones should be measured. The aims of management are to:

- identify and treat any underlying pathology
- ensure normal psychological adaptation to puberty and adulthood
- accelerate growth and promote entry into puberty if necessary.

Following reassurance that puberty will occur, treatment is usually not required. Should treatment be wanted, oral oxandrolone can be used in young males. This weakly androgenic anabolic steroid will induce some catch-up growth but not secondary sexual characteristics. In older boys, low-dose intramuscular testosterone will accelerate growth as well as inducing secondary sexual characteristics. Females may be treated with oestradiol.

Disorders of sexual differentiation (DSD)

The fetal gonad is initially bipotential (Fig. 11.17). In the male, a testis-determining gene on the Y chromosome (*SRY*) is responsible for the differentiation of the gonad into a testis. The production of testosterone and its metabolite, dihydrotestosterone, results in the development of male genitalia. In the absence of *SRY*, the gonads become ovaries and the genitalia female.

Rarely, newborn infants may be born with a disorder of sexual differentiation and there may be uncertainty about the infant's sex. A disorder of sexual differentiation may be secondary to:

- Excessive androgens producing virilisation in a female – the commonest cause of this is congenital adrenal hyperplasia
- Inadequate androgen action, producing undervirilisation in a male – this can result from inability to respond to androgens (a receptor problem – androgen insensitivity syndrome, which may be complete or partial) or to convert testosterone to dihydrotestosterone (5α-reductase deficiency) or abnormalities of the synthesis of androgens from cholesterol
- Gonadotrophin insufficiency, also seen in several syndromes such as Prader–Willi syndrome and congenital hypopituitarism, which results in a small penis and cryptorchidism
- Ovotesticular disorder of sex development (DSD) (previously known as true hermaphroditism), caused by both XX- and Y-containing cells being present in the fetus leading to both testicular and ovarian tissue being present and a complex external phenotype; this is rare.

All parents and their relatives are desperate to know the sex of their newborn baby. However, if the genitalia are abnormal, the infant's sex must not be assigned until detailed assessment by medical, surgical and psychological specialists has been performed followed by full discussion with the parents. Birth registration must be delayed until this has been completed.

Sexuality is complex and depends on more than the phenotype, chromosomes and hormone levels. Before the most appropriate sex of rearing is decided upon, the karyotype needs to be determined, adrenal and sex hormone levels measured, and ultrasound of the internal structures and gonads performed. Sometimes laparoscopic imaging and biopsy of internal structures are necessary. In many disorders of sexual differentiation, it has been usual to raise the child as a female, as it is easier to fashion female external genitalia, whereas it is not possible surgically to create an adequately functioning penis. However, it may be impossible to predict the sexual identity of the child in eventual adult

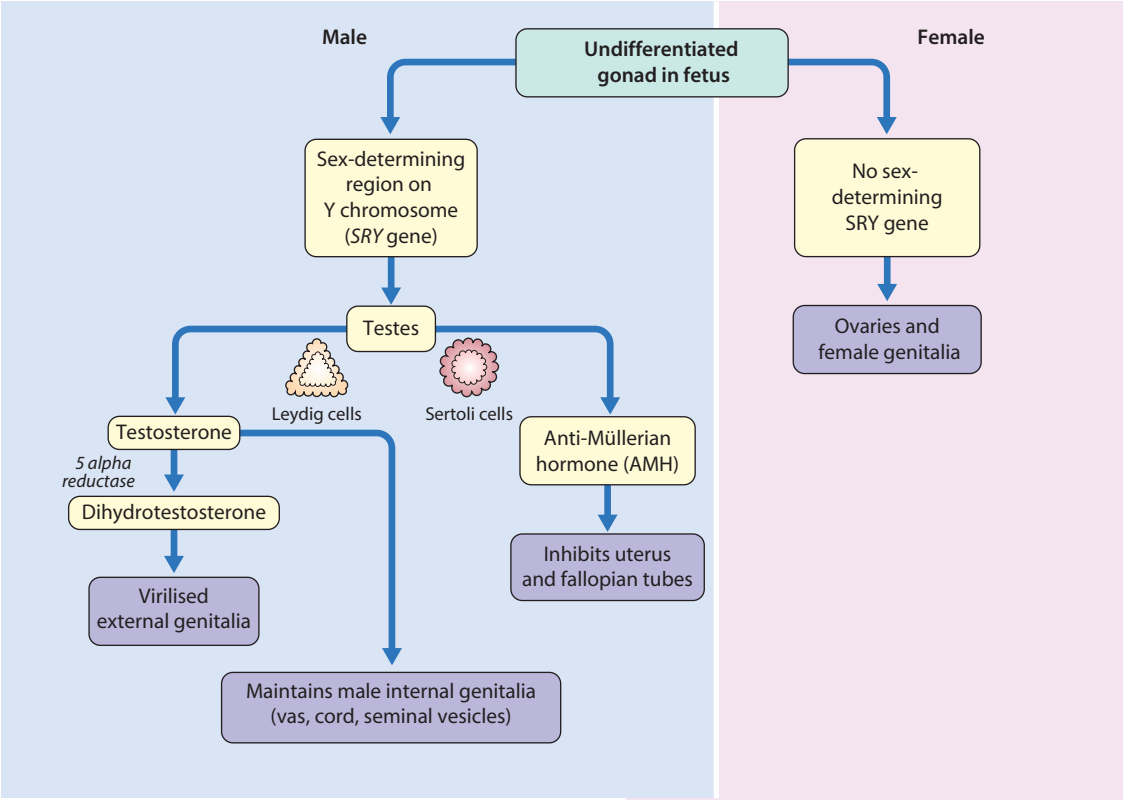


Figure 11.17 Sexual differentiation in the fetus.

life and further support or gender reassignment may be required. For this reason, there is a move toward delaying definitive surgery to allow the affected individual to give informed consent to any reconstructive procedures. This is a controversial area and is best managed by experienced multidisciplinary teams.

If there is abnormal sexual differentiation at birth:

- **Do not guess the infant's gender**
- **The most common cause of a disorder of sexual development is female virilisation from congenital adrenal hyperplasia.**

Congenital adrenal hyperplasia (CAH)

A number of autosomal recessive disorders of adrenal steroid biosynthesis result in congenital adrenal hyperplasia. Its incidence is about 1 in 5000 births, and it is commoner in the offspring of consanguineous marriages. Over 90% have a deficiency of the enzyme 21-hydroxylase, which is needed for cortisol biosynthesis. About 80% are also unable to produce aldosterone, leading to salt loss (low sodium and high potassium) (Fig. 11.18). In the fetus, the resulting cortisol deficiency stimulates the pituitary to produce adrenocorticotrophic hormone (ACTH), which drives overproduction of adrenal androgens.

Presentation:

- Virilisation of the external genitalia in female infants, with clitoral hypertrophy and variable fusion of the labia (see Case History 11.4)
- In the infant male, the penis may be enlarged and the scrotum pigmented, but these changes are seldom identified
- A salt-losing adrenal crisis in the 80% of males who are salt losers; this occurs at 1–3 weeks of age, presenting with vomiting and weight loss, floppiness and circulatory collapse
- Tall stature in the 20% of male non-salt losers; both male and female non-salt losers also develop a muscular build, adult body odour, pubic hair and acne from excess androgen production, leading to precocious pubarche.

There may be a family history of neonatal death if a salt-losing crisis had not been recognised and treated.

Diagnosis

This is made by finding markedly raised levels of the metabolic precursor 17α-hydroxyprogesterone in the blood. In salt losers, the biochemical abnormalities are:

- Low plasma sodium
- High plasma potassium
- Metabolic acidosis
- Hypoglycaemia.

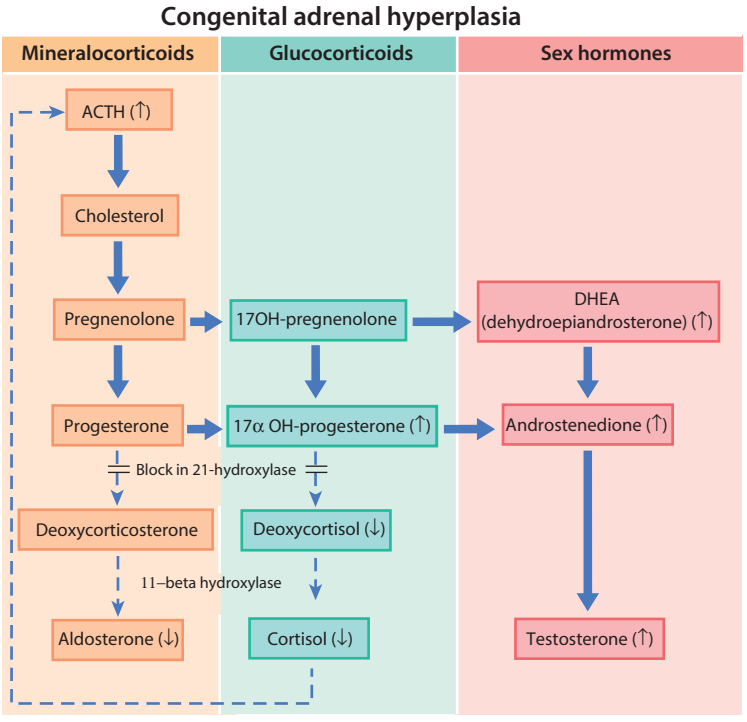


Figure 11.18 Abnormal adrenal steroid biosynthesis from 21-hydroxylase deficiency is the commonest form of congenital adrenal hyperplasia. ACTH, adrenocorticotrophic hormone.

Case History

11.4 Abnormal genitalia at birth

The appearance of this newborn infant's genitalia is shown in Figure 11.19.

Investigation revealed:

- A normal female karyotype, 46XX
- The presence of a uterus on ultrasound examination
- A markedly raised plasma 17α-hydroxyprogesterone concentration, confirming congenital adrenal hyperplasia.
- A low sodium and a high potassium level
- A low bicarbonate (metabolic acidosis) and high urea (dehydration).

Plasma electrolytes identified a low sodium and a high potassium, indicating that the baby had the salt-losing form of CAH. After detailed explanation with her parents, she was started on oral hydrocortisone and fludrocortisone replacement therapy and oral salt replacement (NaCl). Surgery was performed at 9 months of age to reduce clitoral size and separate the labia. Her growth, biochemistry and bone age were monitored frequently at follow-up and she attained normal adult height. Psychological counselling and support were offered around puberty and further genital surgery was needed before she became sexually active.

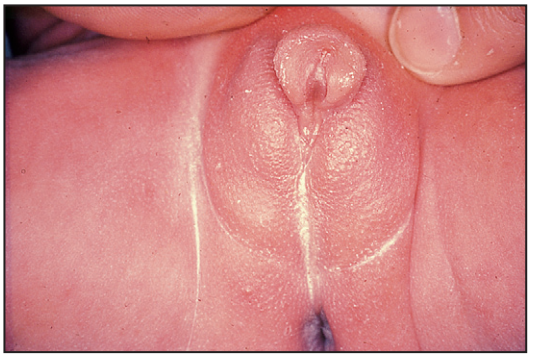


Figure 11.19 Abnormal genitalia at birth. Investigation established that this was a female infant with congenital adrenal hyperplasia causing clitoral hypertrophy with fusion of the labia.

Severe hypospadias and bilateral undescended testes – a male or virilised female? The karyotype and a pelvic ultrasound are required.

Management

Affected females will sometimes require corrective surgery to their external genitalia within the first year but as they have a uterus and ovaries they should usually be reared as girls and are able to have children. Definitive surgical reconstruction is usually delayed until late puberty. Males in a salt-losing crisis require saline, dextrose and hydrocortisone intravenously.

The long-term management of both sexes is with:

- Lifelong glucocorticoids to suppress ACTH levels (and hence testosterone) to allow normal growth and maturation
- Mineralocorticoids (fludrocortisone) if there is salt loss; before weaning, infants may need added sodium chloride
- Monitoring of growth, skeletal maturity and plasma androgens and 17α-hydroxyprogesterone – insufficient hormone replacement results in increased ACTH secretion and androgen excess, which will cause rapid initial growth and skeletal maturation at the expense of final height; excessive hormonal replacement will result in skeletal delay and slow growth
- Additional hormone replacement to cover illness or surgery, as they are unable to mount a cortisol response.

Death can occur from adrenal crisis at the time of illness or injury. Females require surgery to reduce clitoromegaly and a vaginoplasty before sexual intercourse is attempted. Females often experience psychosexual problems, which may relate to the high androgen levels experienced in utero prior to diagnosis.

Prenatal diagnosis and treatment are possible when a couple have had a previously affected child. Dexamethasone may be given to the mother around the time of conception, and continued if the fetus is found to be female, in order to reduce fetal ACTH drive and hence the virilisation.

Summary

Congenital adrenal hyperplasia

- Autosomal recessive disorder of adrenal steroid biosynthesis
- Females present with virilisation of the external genitalia
- Males present with salt loss (80%) or tall stature and precocious puberty (20%)
- Long-term medical management with lifelong glucocorticoids, mineralocorticoids/sodium chloride if salt loss
- Additional corticosteroids to cover illness or surgery
- Salt-losing adrenal crisis needs urgent treatment with hydrocortisone, saline and glucose given intravenously
- Monitor growth, skeletal maturity, plasma androgens and 17α-hydroxyprogesterone
- Surgery for females.

Further reading

Brook C, Clayton P, Brown R: *Brook's Clinical Paediatric Endocrinology*, ed 6, Oxford, 2009, Blackwell.

Lifshitz F, editor: *Pediatric Endocrinology*, ed 5, New York, 2007, Marcel Dekker.

Wales JKH, Rogol AD, Wit JM: *Pediatric Endocrinology and Growth*. London, 2003, Saunders.

This page intentionally left blank

12

Nutrition

The nutritional vulnerability of infants and children 201
Infant feeding 203
Failure to thrive 206
Malnutrition 209
Vitamin D deficiency 213
Vitamin A deficiency 215
Obesity 215
Dental caries 217

Notable features of nutrition in children are:

- The optimal nutrition for newborn infants is from breast-feeding.
- Inadequate nutrition in infants and young children rapidly leads to weight loss followed by growth failure, commonly called failure to thrive, which if severe and prolonged leads to malnutrition
- Whereas malnutrition is a major cause of morbidity and death in developing countries, obesity is the major nutritional problem in developed countries.

The nutritional vulnerability of infants and children

Infants and children are more vulnerable to poor nutrition than are adults. There are a number of reasons for this.

Low nutritional stores

Newborn infants, particularly those born before term, have poor stores of fat and protein (Fig. 12.1). The smaller the child, the less the calorie reserve and the shorter the period the child will be able to withstand starvation.

High nutritional demands for growth

The nourishment children require, per unit body size, is greatest in infancy (Table 12.1), because of their rapid growth during this period. At 4 months of age, 30% of an infant's energy intake is used for growth, but by 1 year of age, this falls to 5%, and by 3 years to 2%. The risk of growth failure from restricted energy intake is therefore greater in the first 6 months of life than in later childhood. Even small but recurrent deficits in early childhood will lead to a cumulative deficit in weight and height.

Rapid neuronal development

The brain grows rapidly during the last trimester of pregnancy and throughout the first 2 years of life. The complexity of interneuronal connections also increases substantially during this time. This process appears to be sensitive to undernutrition. Even modest energy deprivation during periods of rapid brain growth and differentiation is thought to lead to an increased risk of adverse neurodevelopmental outcome. This is not surprising when one considers that at birth the brain accounts for approximately two-thirds of basal

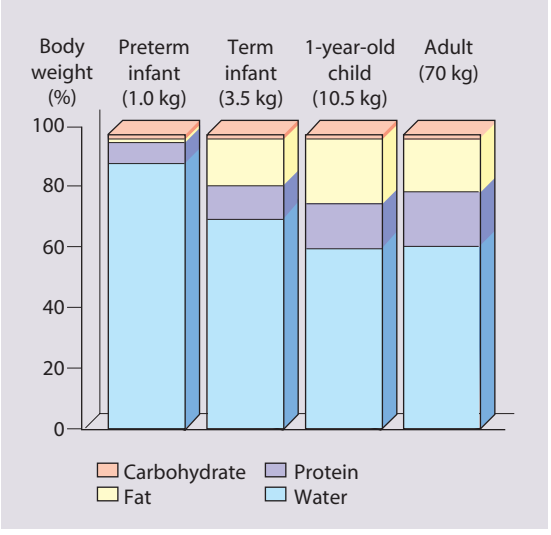


Figure 12.1 Body composition of preterm and term infants, children and adults. Newborn infants, particularly the preterm, have poor stores of fat and protein.

Table 12.1 Reference values for energy and protein requirements

Age	Energy (kcal/kg per 24 h)	Protein (g/kg per 24 h)
0–6 months	115	2.2
6–12 months	95	2.0
1–3 years	95	1.8
4–6 years	90	1.5
7–10 years	75	1.2
Adolescence	(male/female)	
11–14 years	65/55	1.0
15–18 years	60/40	0.8

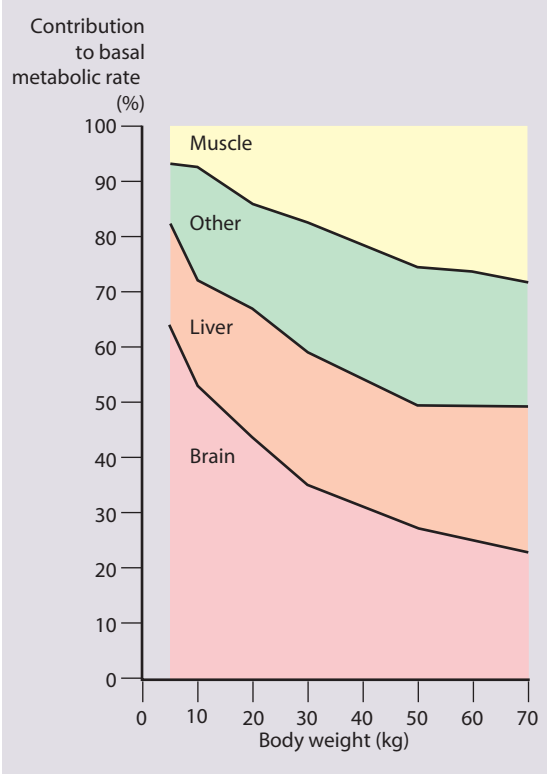


Figure 12.2 The relative contribution to basal metabolic rate derived from brain, liver and muscle changes with growth. Whereas the brain accounts for two-thirds of the basal metabolic rate at birth, this falls to 25% in adults. (Adapted from Halliday MA. 1971. *Pediatrics* 47(1 Suppl 2):169.)

metabolic rate, and at 1 year for about 50% (Fig. 12.2). Many studies have drawn attention to the delayed development seen in children suffering from protein-energy malnutrition due to inadequate food intake, although inadequate psychosocial stimulation may also contribute.

Acute illness or surgery

A child's nutrition may be compromised following an acute illness or surgery. Infants are prone to recurrent infections, which reduce food intake and increase nutritional demands. Following surgery, after a brief anabolic phase, catecholamine secretion is increased, causing the metabolic rate and energy requirement to increase. Urinary nitrogen losses may become so great that it is impossible to achieve a positive nitrogen balance and weight is lost. After uncomplicated surgery, this phase may last for a week, but it can last several weeks after extensive burns, complicated surgery or severe sepsis. Thereafter, previously lost tissue is replaced and a positive energy and nitrogen balance can be achieved. However, infants may not show catch-up growth unless their energy intake is as high as 150–200 kcal/kg per day compared with the normal of 95–115 kcal/kg per day.

Long-term outcome of early nutritional deficiency

Linear growth of populations

Growth and nutrition are closely related, such that the mean height of a population reflects its nutritional status. Thus, in the developed world, people have become taller. Height is adversely affected by lower socioeconomic status and increasing number of children in families. Children's size increases amongst populations emigrating from poor to more affluent countries.

Disease in adult life

Evidence suggests that undernutrition in utero resulting in growth restriction is associated with an increased incidence of coronary heart disease, stroke, non-insulin-dependent diabetes and hypertension in later life (Fig. 12.3). There is also a similar but weaker association with low weight at 1 year of age. The mechanism is unclear, but it is recognised that fetal undernutrition leads to redistribution of blood flow and changes in fetal hormones, such as insulin-like growth factors and cortisol. Alternatively, it may be the rapid, postnatal growth (catch-up) seen in babies suffering from intrauterine growth restriction that is the causal factor.

Summary

Nutritional vulnerability

Infants are more vulnerable to poor nutrition because of:

- Poor stores of fat and protein
- Extra nutritional demands for growth – the weight of a term infant doubles by 4 months and trebles by 1 year.
- More frequent intercurrent illnesses that reduce food intake and increase nutritional demands.

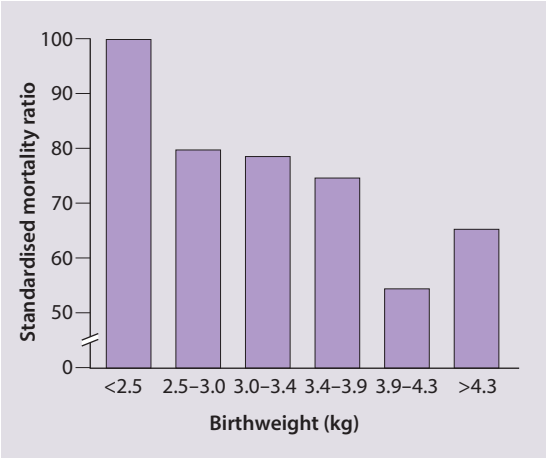


Figure 12.3 Death rates from coronary heart disease according to birthweight showing increased rate in low birthweight babies. (After Barker DJ. 1999. Fetal origins of adult disease. *Growing up in Britain: Ensuring the Healthy Future for our Children.* A Study of 4–5 year olds. BMJ Books, London.)



Figure 12.4 Successful breast-feeding of preterm twins.

Infant feeding

Breast-feeding

There can be no doubt that breastmilk is the best diet for babies, although the popularity of breast-feeding has frequently reflected fashion. The prevalence of breast-feeding in the UK has increased, and 78% of mothers breast-feed their infants at birth. The Department of Health guidelines in the UK, endorsing the World Health Organization recommendation, is that mothers should breast-feed exclusively for the first 6 months of life, though most are weaned to solid food before this age.

Advantages (see Box 12.1)

In developing countries, where the environment is often highly contaminated, breast-feeding dramatically improves survival during infancy as a result of reduced gastrointestinal infection. Consequently, breast-feeding is one of the four most important World Health Organization strategies for improving infant and child survival. The superiority of breast milk over modern adapted cow's milk formulae is less easy to prove in developed countries. This is partly because it is impossible to conduct randomised studies and partly because of confounders such as social class, education and smoking.

However, there is convincing evidence that gastrointestinal infection is less common in breast-fed infants even in developed countries. There is also evidence that human milk feeds reduce the incidence of necrotising enterocolitis in preterm infants.

Many mothers who breast-feed find that it helps them establish an intimate, loving relationship with their baby. However, establishing breast-feeding is not always straightforward, and many mothers need help and encouragement.

Breast-feeding is associated with a reduced incidence of obesity, diabetes mellitus and hypertension in later life. There is also a reduction in breast cancer in mothers who breast-feed.

Exclusive breast-feeding in early infancy is life-saving in developing countries.

Potential complications (Box 12.2)

As one cannot readily tell how much milk a baby is taking from the breast, the baby's weight should be checked regularly, every few days in the first couple of weeks, then weekly until feeding is well established. Successful breast-feeding of twins can be achieved, but is more difficult (Fig. 12.4). It is rarely possible to totally breast-feed triplets and higher-order births. Preterm infants can be breast-fed, but the milk will need to be expressed from the breast until the infant can suck. Maintaining the supply of milk can be a problem for mothers of preterm babies.

While two-thirds of mothers in the UK initially breast-feed, this proportion rapidly declines during the first few months (Fig. 12.5). Nearly 90% of social class I mothers start breast-feeding, but only 60% of mothers from social class V. Breast-feeding is restrictive for the mother, as others cannot take charge of her baby for any length of time. This is particularly important if she goes to work and may delay her return, which may cause financial hardship for the family. Facilities for breast-feeding in public places remain limited. Failure to establish breast-feeding will sometimes cause significant emotional upset in the mother.

Establishing breast-feeding

Colostrum, rather than milk, is produced for the first few days. Colostrum differs from mature milk in that the content of protein and immunoglobulin is much higher. Volumes are low, but water or formula supplements are not required while the supply of breast milk is becoming established.

The first breast-feed should take place as soon as possible after birth. Subsequently, frequent suckling is beneficial as it enhances the secretion of the hormones initiating and promoting lactation (Fig. 12.6).

Box 12.1 Why breast is best – the advantages of breast milk

Advantages of breast-feeding for the infant are that it:

- provides the ideal nutrition for infants during the first 4–6 months of life
- is life-saving in developing countries
- reduces the risk of gastrointestinal infection, and, in preterm infants, of necrotising enterocolitis
- enhances mother–child relationship
- reduces risk of insulin-dependent diabetes, hypertension and obesity in later life.

Advantages for the mother are that it:

- promotes close attachment between mother and baby
- increases the time interval between children, which is important in reducing birth rate in developing countries
- helps with a possible reduction in premenopausal breast cancer.

Scientific explanation of some of the properties of breast milk

Anti-infective properties

Humoral

Secretory IgA	Comprises 90% of immunoglobulin in human milk. Provides mucosal protection, but of uncertain benefit
Bifidus factor	Promotes growth of *Lactobacillus bifidus*, which metabolises lactose to lactic and acetic acids. The resulting low pH may inhibit growth of gastrointestinal pathogens
Lysozyme	Bacteriolytic enzyme
Lactoferrin	Iron-binding protein. Inhibits growth of *Escherichia coli*
Interferon	Antiviral agent

Cellular

Macrophages	Phagocytic. Synthesise lysozyme, lactoferrin, C3, C4
Lymphocytes	T cells may transfer delayed hypersensitivity responses to infant. B cells synthesise IgA

Nutritional properties

Protein quality	More easily digested curd (60:40 whey:casein ratio)
Lipid quality	Rich in oleic acid (with palmitate in C-2 position). Improved digestibility and fat absorption Enhanced lipolysis lipase.
Calcium: phosphorus ratio of 2:1	Prevents hypocalcaemic tetany and improves calcium absorption
Renal solute load	Low
Iron content	Bioavailable (40–50% absorption)
Long-chain polyunsaturated fatty acids	Structural lipids; important in retinal development

Box 12.2 Potential complications of breast-feeding

Unknown intake	Volume of milk intake not known
Transmission of infection	Maternal CMV, hepatitis B and HIV – increases risk of transmission to the baby
Breast-milk jaundice	Mild, self-limiting, unconjugated hyperbilirubinaemia; continue breast-feeding
Transmission of drugs	Antimetabolites and some other drugs contra-indicated. Check formulary
Nutrient inadequacies	Breast-feeding beyond 6 months without timely introduction of appropriate solids may lead to poor weight gain and rickets
Vitamin K deficiency	Insufficient vitamin K in breast milk to prevent haemorrhagic disease of the newborn. Supplementation is required
Potential transmission of environmental contaminants	Nicotine, alcohol, caffeine, etc.
Less flexible	Other family members cannot help or take part. More difficult in public places
Emotional upset	If difficulties or lack of success can be upsetting

Primates probably do not breast-feed instinctively. Monkeys bred in captivity in zoos have to be taught how to breast-feed by their keepers. It is therefore important that breast-feeding should have as high a public profile as possible. Women who have never seen an infant being breast-fed are less likely to want to breast-feed themselves. Education in schools and during pregnancy about the advantages of breast-feeding is advantageous. Advice and support from other women who have breast-fed may be important in dealing with early problems such as engorgement or cracked nipples.

Newborn infants of mothers planning to breast-feed should ideally not be given any formula feeds.

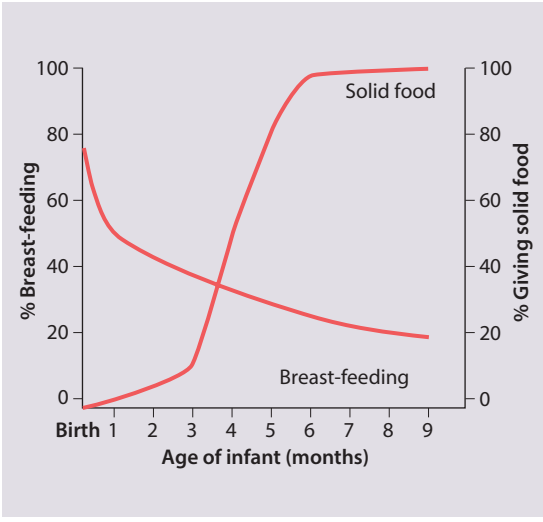


Figure 12.5 Prevalence of breast-feeding and proportion of infants given solid feeds during the first 9 months of life in the UK. Over the last 10 years there has been a marked increase in the prevalence of breast-feeding and delay in weaning onto solid food (Infant Feeding Survey, 2005).

Formula-feeding

Infants who are not breast-fed require a formulafeed based on cow's milk. Unmodified cow's milk is unsuitable for feeding in infancy as it contains too much protein and electrolytes and inadequate iron and vitamins. Even after considerable modification, differences remain between formula feeds and breastmilk (Table 12.2)

All milks currently available in the UK have been modified to make their mineral content and renal solute load comparable with that of mature human milk. Since these changes were introduced in the UK (in the 1970s), there has been an impressive reduction in the incidence of hypernatraemic dehydration in infants with gastroenteritis. There is no evidence that any one of the many brands is superior to any other.

Introduction of whole, pasteurised cow's milk

Breast-feeding or formula-feeding is recommended until the age of 12 months, and there are advantages in continuing to 18 months of age. Pasteurised cow's milk may be given from 1 year of age but is deficient

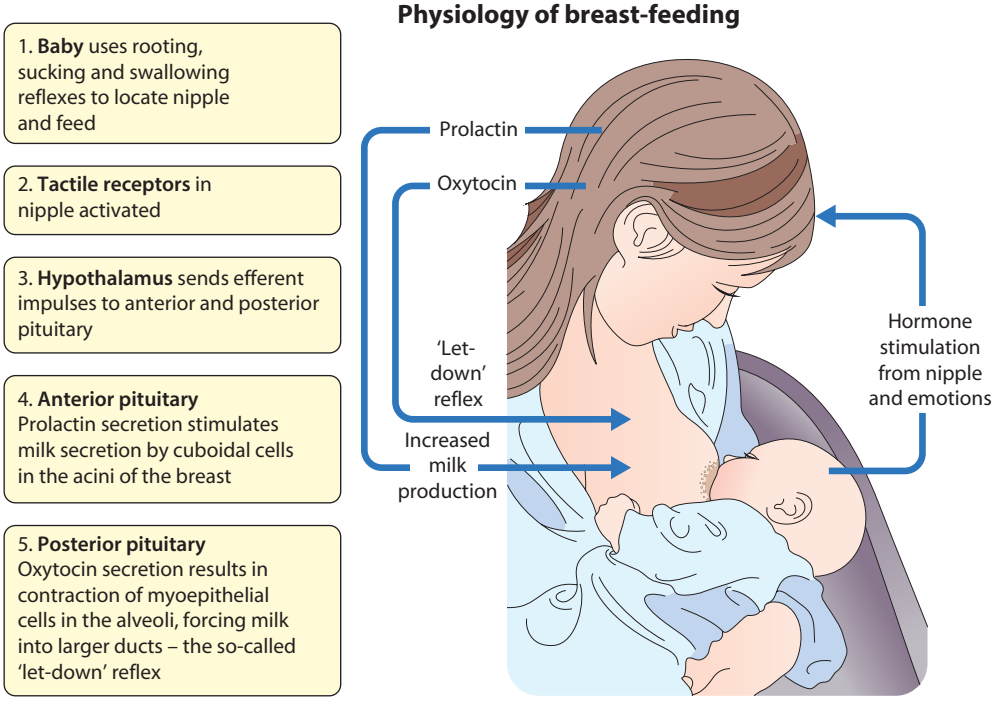


Figure 12.6 Physiology of breast-feeding.

Table 12.2 A comparison of human milk, cow's milk and infant formula (per 100 ml)

	Mature breast milk	Cow's milk	Infant formula (modified cow's milk)
Energy (kcal)	62	67	60–65
Protein (g)	1.3	3.5	1.5–1.9
Carbohydrate (g)	6.7	4.9	7.0–8.6
Casein : whey	40 : 60	63 : 37	40 : 60 to 63 : 37
Fat (g)	3.0	3.6	2.6–3.8
Sodium (mmol)	0.65	2.3	0.65–1.1
Calcium (mmol)	0.88	3.0	0.88–2.1
Phosphorus (mmol)	0.46	3.2	0.9–1.8
Iron (μmol)	1.36	0.9	8–12.5

in vitamins A, C and D and in iron, and supplementation will be required unless the infant is having a good diet of mixed solids. Alternatively, 'follow-on' formulae can be used. They contain more protein and sodium than infant formulae and, in contrast to cow's milk, are fortified with iron and vitamins. Children on cow's milk should receive full fat milk up to the age of 5 years.

Infant formula, not unmodified cow's milk, should be used in the first year of life.

Specialised infant formula

A specialised formula may be used for cow's milk protein allergy/intolerance, lactose intolerance (primary lactase deficiency or post-gastroenteritis intolerance), cystic fibrosis, neonatal cholestatic liver disease and following neonatal intestinal resection.

In a cow's milk-based formula, the protein is derived from cow's milk protein, the carbohydrate is lactose and the fat mainly long-chain triglycerides. In a specialised formula, the protein is either hydrolysed cow's milk protein, amino acids or from soya, the carbohydrate is glucose polymer and the fat a combination of medium- and long-chain triglycerides. Medium-chain triglycerides are directly absorbed into small intestine and need neither pancreatic enzymes nor bile salts for this process.

A soya formula should not be used below 6 months of age as it has a high aluminium content and contains phytoestrogens (plant substances that mimic the effects of endogenous oestrogens). There is no compelling evidence that the use of a specialised formula prevents the development of atopy (eczema, asthma, etc.).

Weaning

Solid foods are recommended to be introduced after 6 months of age, although they are often introduced earlier as parents often consider that their infant is hungry. It is done gradually, initially with small quantities of pureed fruit, root vegetables, or rice. If weaning takes place before 6 months of age, wheat, eggs and fish should be avoided. Foods high in salt and sugar should also be avoided and honey should not be given until 1 year of age because of risk of infantile botulism. After 6 months of age, breastmilk becomes increasingly nutritionally inadequate as a sole feed, as it does not provide sufficient energy, vitamins or iron.

Failure to thrive

The term 'failure to thrive' is used to describe suboptimal, weight gain in infants and toddlers. It may also be referred to as weight or growth faltering in case parents consider the term critical of their care. Recognition of the entity depends upon demonstration of inadequate weight gain when plotted on a centile chart, mild failure to thrive being a fall across two centile lines and severe being a fall across three centile lines. Between 6 weeks and 1 year of age, only 5% of children will cross two lines, and only 1% will cross three. Most children with 'failure to thrive' have a weight below the 2nd centile. However, the weight of some children with failure to thrive, i.e. they are failing to gain or are losing weight, may still be above the 2nd centile. Repeated observations are therefore essential and are usually available from the child's personal child health record. A single observation of weight is difficult to interpret unless markedly discrepant from the head circumference or length, although the further the weight is below the 2nd centile, the more likely the child is 'failing to thrive'. A weight below the 0.4th centile should always trigger an evaluation.

Differentiating the infant who is failing to thrive from a normal but small or thin baby is often a problem (Fig. 12.7). Normal but short infants have no symptoms, are alert, responsive and happy, and their development is satisfactory. The parents may be short (low mid-parental height) or the infant may have been extremely preterm or growth-restricted at birth. Any intercurrent illness may be accompanied by a temporary failure to gain weight.

An additional diagnostic problem is 'catch-down' (as opposed to 'catch-up') weight. This is when an infant's weight falls from the birth centile, which is

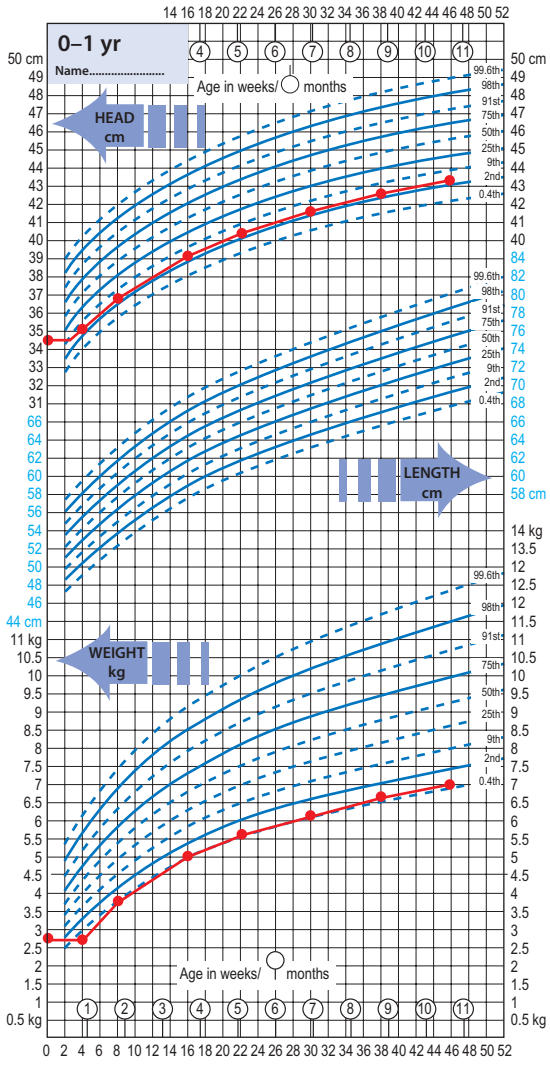


Figure 12.7 Growth chart showing normal weight gain and growth in a constitutionally small infant. The further below the 2nd, and especially the 0.4th, centile, the more likely it is that there will be an organic cause. (Chart © RCPCH, WHO, Department of Health.)

determined by the intrauterine environment, to a lower, genetically determined growth centile. These infants need only close monitoring of their growth over a few months.

While children with recent-onset failure to thrive usually maintain their height, this may become compromised by prolonged, inadequate weight gain. The child's developmental progress may also be adversely affected.

Causes

Failure to thrive is usually classified as organic or non-organic (Fig. 12.8). Traditionally non-organic failure to thrive is believed to be associated with a broad spectrum of psychosocial and environmental deprivation. It is estimated that 5–10% of children with failure to thrive will be on a child protection register or be subjected to abuse or neglect, while in a larger proportion, socioeconomic deprivation is an important contributing factor.

The mother may be depressed, have an eating disorder herself or have poor understanding of her baby's needs. There may be poor housing, poverty, inadequate social support and lack of an extended family, which make good child care even more difficult. However, some studies suggest that failure to thrive is not more common in deprived than in non-deprived communities, and that identification of deprivation leads to the inappropriate application of that diagnostic label. Undernutrition is the final common pathway for poor weight gain in most cases of organic and non-organic failure to thrive, and in many cases both organic and environmental factors are present.

Organic causes are listed in Figure 12.8. Less than 5% of children with failure to thrive will be found to have an organic cause.

Clinical features and investigation

Studying the growth chart in combination with the history and examination of the child is key to its evaluation. The history should focus on:

- A detailed dietary history, including a food diary over several days
- Feeding, including details of exactly what happens at mealtimes
- Is the child well with lots of energy or does the child have other symptoms such as diarrhoea, vomiting, cough, lethargy?
- Was the child premature or had intrauterine growth restriction at birth or any significant medical problems?
- The growth of other family members and any illnesses in the family
- Is the child's development normal?
- Are there psychosocial problems at home?

Examination should focus on identifying signs of organic disease – dysmorphic features, signs suggestive of malabsorption (distended abdomen, thin buttocks, misery), signs suggestive of chronic respiratory disease (chest deformity,clubbing), signs of heart failure and evidence of nutritional deficiencies.

Further information about the child and family from the health visitor, general practitioner or other professionals involved with the family can be particularly helpful. Investigations to be considered are listed in Box 12.3.

In some children who are failing to thrive, a full blood count and serum ferritin may be helpful to identify iron deficiency anaemia. This is usually secondary to inadequate iron intake and correcting it may improve appetite. In most instances, no investigations are required.

Management

The management of most non-organic failure to thrive is multidisciplinary and is carried out in primary care. The health visitor is well placed to make home visits to

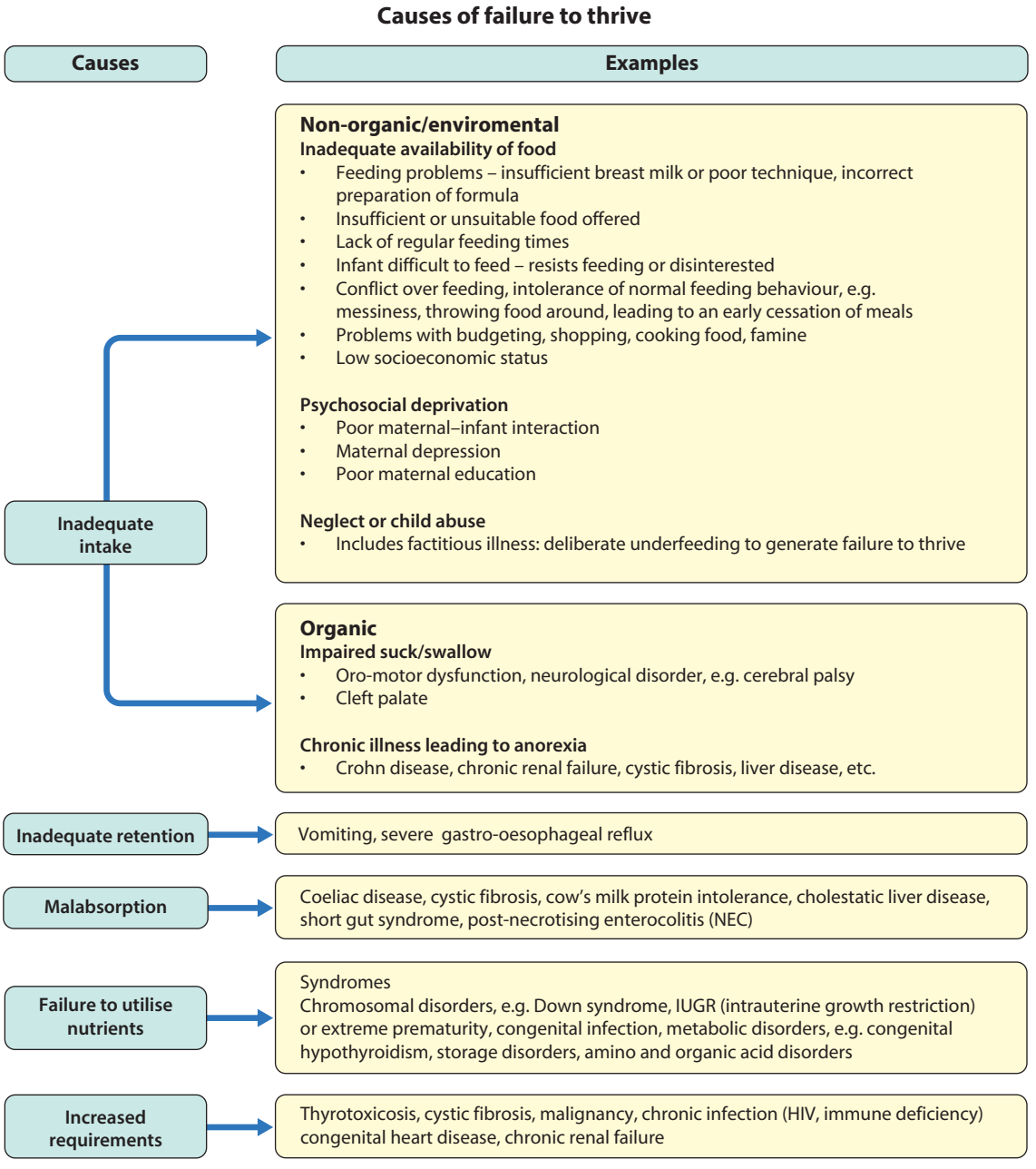


Figure 12.8 The main causes of failure to thrive.

assess eating behaviour and provide support. Direct practical advice following observation may well be beneficial. A paediatric dietician may be helpful in assessing the quantity and composition of food intake, and recommending strategies for increasing energy intake and a speech and language therapist has specialist skills with feeding disorders. Input from a clinical psychologist and from social services may also be appropriate. Nursery placement may be helpful in alleviating stress at home and assist with feeding.

Hospital admission is usually only necessary in children under 6 months with severe failure to thrive, requiring active refeeding. While hospital admission may offer the opportunity to observe and improve the mother's method and skill in feeding, this rarely transfers from the artificial hospital environment to home. In extreme cases, hospital admission can be used to demonstrate that the child will gain weight when fed appropriately.

Outcome

Follow-up studies suggest that children with non-organic failure to thrive continue to under-eat (see Case History 12.1). Although there is usually a gradual improvement in the preschool years, a lasting deficit is common and these children tend to remain underweight. In contrast, impairment of development is only short term.

Box 12.3 Investigations to be considered in 'failure to thrive'

Investigation	Significance of an abnormality
Full blood count and differential white cell count	Anaemia, neutropenia, lymphopenia (immune deficiency)
Serum creatinine urea, electrolytes, acid–base status, calcium, phosphate	Renal failure, renal tubular acidosis, metabolic disorders, William syndrome
Liver function tests	Liver disease, malabsorption, metabolic disorders
Thyroid function tests	Hypothyroidism or hyperthyroidism
Acute phase reactant, e.g. C-reactive protein	Inflammation
Ferriaztin	Iron deficiency anaemia
Immunoglobulins	Immune deficiency
IgA tissue transglutaminase antibodies	Coeliac disease
Urine microscopy, culture and dipsticks	Urinary tract infection, renal disease
Stool microscopy, culture and elastase	Intestinal infection, parasites, elastase decreased in pancreatic insufficiciency
Karyotype in girls	Turner syndrome
Chest X-ray and sweat test	Cystic fibrosis

Summary

Failure to thrive

- is a description, not a diagnosis
- weights of infants are only helpful if accurate and plotted on a centile chart
- is present if an infant's weight falls across two centile lines
- is likely to be present the further the weight is below the 2nd centile
- is mostly due to inadequate food intake
- is accompanied by abnormal symptoms or signs if there is organic disease
- most affected infants and toddlers do not require any investigations and are managed in primary care by increasing energy intake by dietary and behavioural modification and monitoring growth.

Malnutrition

Worldwide, malnutrition is common and is responsible directly or indirectly for about a third of all deaths of children under 5 years of age. Primary malnutrition also continues to occur in developed countries as a result of poverty, parental neglect or poor education. Specific nutritional deficiencies, particularly of iron, remain common in developed countries. Restrictive diets may be iatrogenic as a result of exclusion diets or parental food fads, or may be self-inflicted.

Malnutrition also occurs in 20–40% of children in specialist children's hospitals. At particular risk are those with chronic illness: e.g. the preterm, congenital heart disease, malignant disease during chemotherapy or bone marrow transplantation, chronic gastrointestinal conditions such as short gut syndrome following extensive bowel resection or inflammatory bowel disease, chronic renal failure or cerebral palsy. Malnutrition results from a combination of anorexia, malabsorption and increased energy requirements because of infection or inflammation. Malnutrition in older children and adolescents may also result from eating disorders.

Assessment of nutritional status

Malnutrition must be recognised and accurately defined for rational decisions to be made about refeeding. Evaluation is divided into assessment of past and present dietary intake, anthropometry and laboratory assessments (Fig. 12.10).

Dietary assessment

Parents are asked to record the food the child eats during several days. This gives a guide to food intake.

Anthropometry

In addition to weight and height, skinfold thickness of the triceps reflects subcutaneous fat stores and can be assessed by measuring it. While it is difficult to measure skinfold thickness accurately in young children, mid-upper arm circumference, which is related to skeletal muscle mass, can be measured easily and repeatedly and is independent of age in children 6 months to 6 years. It is especially useful for screening children for malnutrition in the community.

Laboratory investigations

These are useful in the detection of early physiological adaptation to malnutrition, but clinical history, examination and anthropometry are of greater value than any single biochemical or immunological measurement.

Case History

12.1 Non-organic failure to thrive

Jamie, aged 11 months, was causing concern to his health visitor as he was not putting on any weight (Fig. 12.9). She arranged for him to be assessed by his general practitioner, who found that he was otherwise well. His mother was a single parent who left school at 16 years and had Jamie at the age of 18. They lived in a high-rise flat and Jamie's mother received income support. Her own mother lived on the other side of the city.

On visiting the home, the health visitor found Jamie's mother to be tense and anxious. In particular, she was worried about making ends meet. She fed Jamie the same food as she ate herself, together with pasteurised milk, which she had started at 6 months of age. The meals were chaotic. After a few mouthfuls, Jamie stopped eating and his mother did not coax him but became frustrated and angry.

Jamie's health visitor suggested strategies for increasing Jamie's food intake (Box 12.4). She continued to provide support and encouragement to his mother and arranged a nursery placement for Jamie. By 2 years of age, he had caught up by one centile line, but still ate erratically.

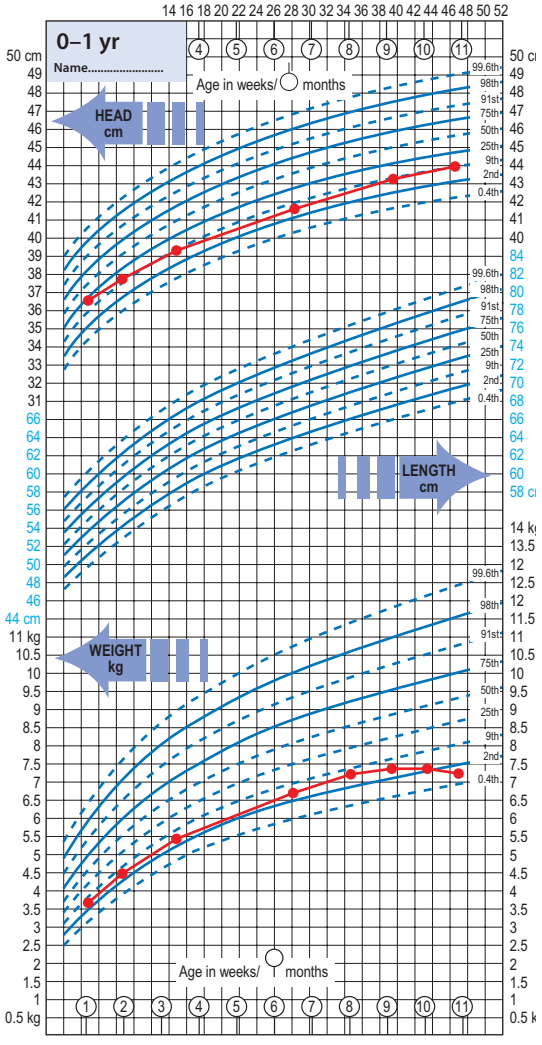


Figure 12.9 Jamie's growth chart. (Chart © RCPCH, WHO, Department of Health.)

Box 12.4 Strategies for increasing energy intake

Dietary

- Three meals and two snacks each day
- Increase number and variety of foods offered
- Increase energy density of usual foods (e.g. add cheese, margarine and cream)
- Decrease fluid intake, particularly squash

Behavioural

- Have meals at regular times, eaten with other family members
- Praise when food is eaten
- Gently encourage child to eat, but avoid conflict
- Never force-feed.

After Wright CM. 2000. Identification and management of failure to thrive: a community perspective. *Archives of Disease in Childhood* 82:5–9.

Consequences of malnutrition

Malnutrition is a multisystem disorder. When severe, immunity is impaired, wound healing is delayed and operative morbidity and mortality increased. Malnutrition worsens the outcome of illness, e.g. respiratory muscle dysfunction may delay a child being weaned from mechanical ventilation. Malnourished children are less active, less exploratory and more apathetic. These behavioural abnormalities are rapidly reversed with proper feeding, but prolonged and profound malnutrition can cause permanent delay in intellectual development.

The role of intensive nutritional support

Children with chronic disorders who are malnourished will grow better if given supplemental nutritional support, which may be provided by the enteral or parenteral route.

Enteral nutrition

Enteral nutrition is used when the digestive tract is functioning, as it maintains gut function, and is safe. Feeds are given nasogastrically, by gastrostomy or

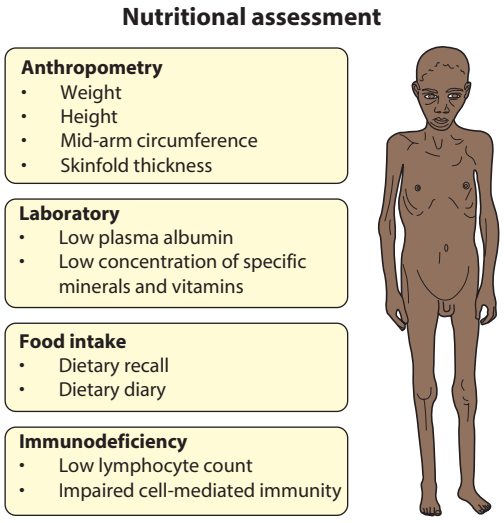


Figure 12.10 Assessment of nutritional status. This cannot be determined by a single measurement but is a composite of a number of variables.

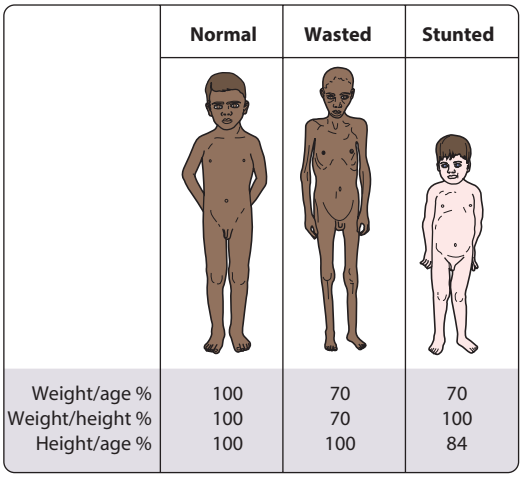


Figure 12.11 Comparison of a normal, wasted and stunted child at 1 year. Low weight for height reveals a child of normal height, but who is thin and wasted, whereas low height for age reveals a short, non-wasted child.

occasionally via a feeding tube in the jejunum. Feeds are often given continuously overnight, allowing the child to feed normally during the day. Gastrostomies can either be created endoscopically or surgically. If long-term supplemental enteral nutrition is required, a gastrostomy is prefered as it avoids repeated replacements of nasogastric tubes, which is distressing for the child.

Parenteral nutrition

Parenteral nutrition can be used exclusively or as an adjunct to enteral feeds to maintain and/or enhance nutrition. The aim is to provide a nutritionally complete feed in an appropriate volume of intravenous fluid. It can be delivered at home. However, it is a complex and expensive form of therapy, requiring a multidisciplinary approach incorporating the skills not only of medical and nursing staff but also pharmacists and dieticians. Short term, it is possible to deliver it via peripherally sited canulae; long term, it is delivered via a central venous catheter as this allows infusion of hyperosmolar solutions and does not require repeated resiting of the cannula. It may be inserted surgically or under radiological guidance. Complications include catheter sepsis or blockage, problems of vascular access on repeated line placement and liver disease from the parenteral nitrition itself.

Marasmus and kwashiorkor

Globally, over one-third of childhood deaths are attributable to undernutrition, which leaves the child susceptible and unable to survive common infections.

The World Health Organization recommends that nutritional status is expressed as:

- Weight for height – a measure of wasting and an index of acute malnutrition (Fig. 12.11). Severe malnutrition is defined as a weight for height more than –3 standard deviations below the median plotted on the WHO standard growth chart. This corresponds to a weight for height <70% below the median.
- Mid upper-arm circumference (MUAC) – <115 mm is severe malnutrition
- Height for age – a measure of stunting and an index of chronic malnutrition

Severe protein-energy malnutrition in children usually leads to marasmus, with a weight for height more than –3 standard deviations below the median, corresponding to <70% weight for height, and a wasted, wizened appearance (Fig. 12.12). Oedema is not present. Skinfold thickness and mid-arm circumference are markedly reduced, and affected children are often withdrawn and apathetic.

Kwashiorkor is another manifestation of severe protein malnutrition (Fig. 12.13), in which there is generalised oedema as well as severe wasting. Because of the oedema, the weight may not be as severely reduced. In addition, there may be:

- a 'flaky-paint' skin rash with hyperkeratosis (thickened skin) and desquamation
- a distended abdomen and enlarged liver (usually due to fatty infiltration)
- angular stomatitis
- hair which is sparse and depigmented
- diarrhoea, hypothermia, bradycardia and hypotension

Malnutrition



Figure 12.12 Marasmus in a 3-month-old baby who was unable to establish breast-feeding because of a cleft palate.



Figure 12.13 Kwashiorkor, a particular manifestation of severe protein-energy malnutrition in some developing countries where infants are weaned late from the breast and the young child's diet is high in starch. There is oedema around the eyes and feet and legs, hyperkeratosis and depigmentation of the skin and redness of the hair.

- low plasma albumin, potassium, glucose and magnesium.

It is unclear why some children with protein-energy malnutrition develop kwashiorkor and others develop marasmus. Kwashiorkor is a feature of children reared in traditional, polygamous societies, where infants are not weaned from the breast until about 12 months of age. The subsequent diet tends to be relatively high in starch. Kwashiorkor often develops after an acute intercurrent infection, such as measles or gastroenteritis. There is some evidence that kwashiorkor is a manifestation of primary protein deficiency with energy intake relatively well maintained or, alternatively, that it results from excess generation of free radicals.

Management

Severe acute malnutrition has a high mortality; about 30% in children require hospital care. In addition to protein and energy deficiency, there is electrolyte and mineral deficiency (potassium, zinc, magnesium) as well as micronutrient and vitamin deficiency (vitamin A).

Acute management includes:

- Hypoglycaemia – common; correct urgently, particularly if coma or severely ill.
- Hypothermia – wrap, especially at night.
- Dehydration – correct, but avoid being overzealous with intravenous fluids, as may lead to heart failure.
- Electrolytes – correct deficiencies, especially potassium.
- Infection – give antibiotics; fever and other signs may be absent. Treat oral candida if present.
- Micronutrients – give vitamin A and other vitamins
- Initiate feeding – small volumes, frequently, including through the night.

Children with no appetite or medical complications need hospital care. Otherwise care can be community based. Although protein deficient, diet is initially low in protein as high protein feeds are not tolerated. Too rapid feeding may result in diarrhoea. Specialised feeds are widely available. Initially Formula 75 (75 kcal/100 ml), subsequently Formula 100 (100 kcal/100 ml) or Ready-to-Use Therapeutic Food (RUTF). During recovery phase, growth is monitored, sensory stimulation should be provided and discharge preparation undertaken.

Summary

Malnutrition

- Worldwide – contributes to about a third of all childhood deaths; often a consequence of war and social disruption, as well as famine and natural disasters
- In developed countries – results from poverty, parental neglect or poor education, restrictive diets, children with feeding disorders or chronic illness or anorexia nervosa
- Can be identified by anthropometric measurement; laboratory tests are not usually required
- Marasmus – weight for height more than 3 Standard Deviations below the median; wasted, wizened appearance; apathetic
- Kwashiorkor – generalised oedema, sparse and depigmented hair, skin rash, angular stomatitis, distended abdomen and enlarged liver, diarrhoea.

Vitamin D deficiency

Vitamin D deficiency usually results from deficient intake or defective metabolism of vitamin D, causing a low serum calcium (Fig. 12.14). This triggers the secretion of parathyroid hormone and normalises the serum calcium but demineralises the bone. Parathyroid hormone causes renal losses of phosphate and consequently low serum phosphate levels, further reducing the potential for bone calcification.

Vitamin D deficiency usually presents with bony deformity and the classical picture of rickets. It can also present without bone abnormalities but with symptoms of hypocalcaemia, i.e. seizures, neuromuscular irritability (tetany), apnoea, stridor. This presentation is more common before 2 years of age and in adolescence, when a high demand for calcium in rapidly growing bone results in hypocalcaemia before rickets develops.

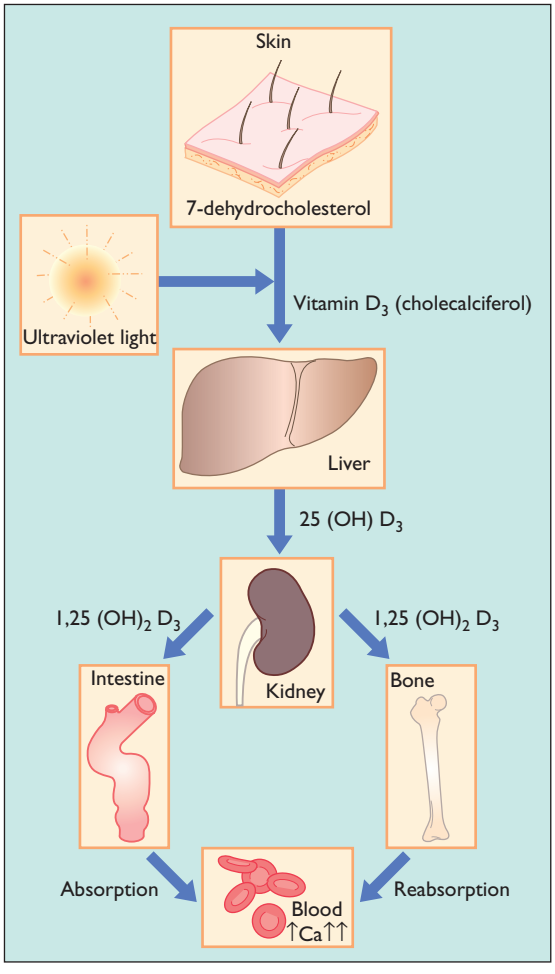


Figure 12.14 Vitamin D metabolism. In most countries, sunlight is the most important source of vitamin D. Vitamin D is not abundant naturally in food, except in fish liver oil, fatty fish and egg yolk. Vitamin D_2 (ergocalciferol) is the form used to fortify food such as margarine. Vitamin D_3 is hydroxylated in the liver and again in the kidney to produce 1,25-dihydroxyvitamin D ($1,25(OH)_2D_3$), the most active form of the vitamin. It is produced following parathyroid hormone secretion in response to a low plasma calcium.

Rickets

Rickets signifies a failure in mineralisation of the growing bone or osteoid tissue. Failure of mature bone to mineralise is osteomalacia.

Aetiology

The causes of rickets are listed in Box 12.5 The predominant cause of rickets during the early twentieth century was nutritional vitamin D deficiency due to inadequate intake or insufficient exposure to direct sunlight. Nutritional rickets still remains the major cause in developing countries. In developed countries, nutritional rickets has become rare, as formula milk and many foods such as breakfast cereals are supplemented with vitamin D. However, nutritional rickets has re-emerged in developed countries in black or Asian infants totally breast-fed in late infancy. It is also seen in extremely preterm infants from dietary deficiency of phosphorus, together with low stores of calcium and phosphorus.

Children with malabsorptive conditions such as cystic fibrosis, coeliac disease and pancreatic insufficiency can develop rickets due to deficient absorption of vitamin D, calcium or both. Drugs, especially anticonvulsants such as phenobarbital and phenytoin, interfere with the metabolism of vitamin D and may also cause rickets. Rickets may also result from impaired metabolic conversion or activation of vitamin D (hepatic and renal disease).

Clinical manifestations

The earliest sign of rickets is a ping-pong ball sensation of the skull (craniotabes) elicited by pressing firmly over the occipital or posterior parietal bones. The costochondral junctions may be palpable (rachitic rosary), wrists (especially in crawling infants) and ankles (especially in walking infants) may be widened and there may be a horizontal depression on the lower chest corresponding to attachment of the softened ribs and with the diaphragm (Harrison sulcus) (Figs 12.15, 12.16). The legs may become bowed (see Fig. 12.15). The clinical features are listed in Box 12.6 (see also Case History 12.2).

Diagnosis

This is made from:

- Dietary history for vitamin and calcium intake
- Blood tests – serum calcium is low or normal, phosphorus low, plasma alkaline phosphatase activity greatly increased, 25-hydroxyvitamin D may be low and parathyroid hormone elevated.
- X-ray of the wrist joint – shows cupping and fraying of the metaphyses and a widened epiphyseal plate.

Box 12.5 Causes of rickets

Nutritional (primary) rickets – risk factors

- Living in northern latitudes
- Dark skin
- Decreased exposure to sunlight, e.g. in some Asian children living in the UK
- Maternal vitamin D deficiency
- Diets low in calcium, phosphorus and vitamin D, e.g. exclusive breast-feeding into late infancy or, rarely, toddlers on unsupervised 'dairy-free' diets
- Macrobiotic, strict vegan diets
- Prolonged parenteral nutrition in infancy with an inadequate supply of parenteral calcium and phosphate

Intestinal malabsorption

- Small bowel enteropathy (e.g. coeliac disease)
- Pancreatic insufficiency (e.g. cystic fibrosis)
- Cholestatic liver disease
- High phytic acids in diet (e.g. chapattis)

Defective production of $25(OH)D_2$

- Chronic liver disease

Increased metabolism of $25(OH)D_3$

- Enzyme induction by anticonvulsants (e.g. phenobarbital)

Defective production of $1,25(OH)_2D_3$

- Hereditary type I vitamin D-resistant (or dependent) rickets (mutation which abolishes activity of renal hydroxylase)
- Familial (X-linked) hypophosphataemic rickets (renal tubular defect in phosphate transport)
- Chronic renal disease
- Fanconi syndrome (renal loss of phosphate)

Target organ resistance to $1,25(OH)_2D_3$

- Hereditary vitamin D-dependent rickets type II (due to mutations in vitamin D receptor gene).

Rickets

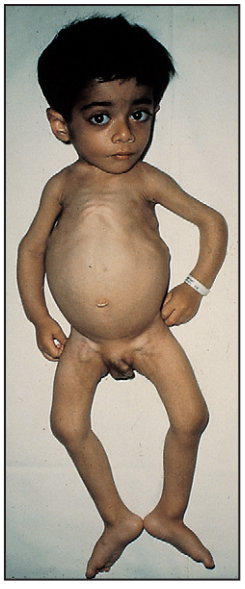


Figure 12.15 Rickets in a 3-year-old boy secondary to coeliac disease. He has frontal bossing, a Harrison sulcus and bow legs.

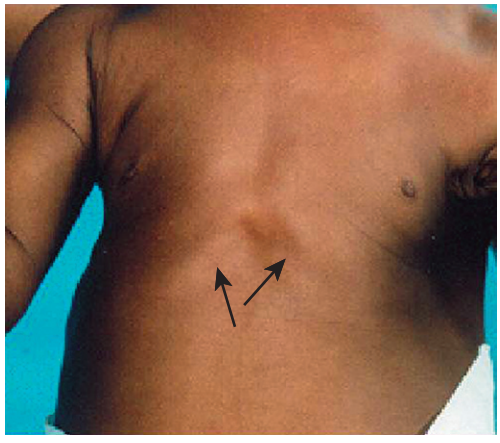


Figure 12.16 Harrison sulcus, indentation of the softened lower ribcage at the site of attachment of the diaphragm. (Courtesy of Dr Nick Shaw.)

Box 12.6 Clinical features of rickets

- Misery
- Failure to thrive/short stature
- Frontal bossing of skull
- Craniotabes
- Delayed closure of anterior fontanelle
- Delayed dentition
- Rickety rosary
- Harrison sulcus (Fig. 12.16)
- Expansion of metaphyses (especially wrist)
- Bowing of weight-bearing bones
- Hypotonia
- Seizures (late).

Case History

12.2 Seizures and rickets

Mohammed, a 13-month-old Somalian boy, was admitted to the A&E department with a generalised afebrile seizure. This was initially controlled with per rectum diazepam. Some 20 minutes later he had another generalised seizure and needed intravenous anticonvulsant to control his seizure.

His mother said that he was a healthy child. He was born at term, birthweight 3.1 kg, and was still breast-fed. Some weaning foods were started at 7–8 months, but he preferred feeding at the breast. He had only recently begun to sit without support.

His weight and head circumference were on the 2nd–9th centile. He had marked frontal bossing, widened wrist (Fig. 12.17) and other epiphyses, Harrison sulci, wide anterior fontanelle, craniotabes and a rachitic rosary. He would not take his weight on standing.

Investigations showed a low calcium and phosphate level, a high alkaline phosphatase and parathyroid hormone level and a very low vitamin D level, confirming rickets. Liver and renal function tests were normal and coeliac screen was negative. His wrist X-ray showed characteristic features (Fig. 12.18). A detailed dietetic history revealed a diet deficient in calcium and vitamin D, confirming nutritional rickets as the cause.

Dietetic input was provided. He was started on oral vitamin D and his solid food intake was increased to ensure that he was receiving sufficient calcium and vitamin D in his diet. His rickets resolved.

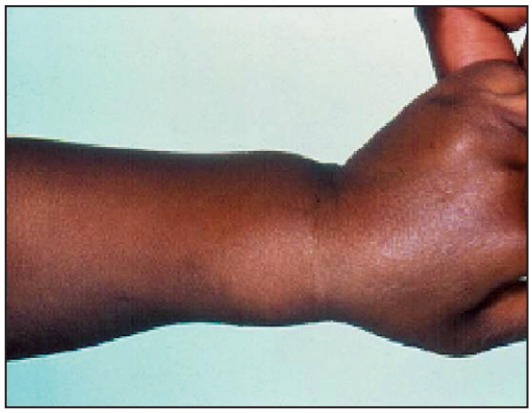


Figure 12.17 Wrist expansion from rickets. (Courtesy of Dr Nick Shaw.)

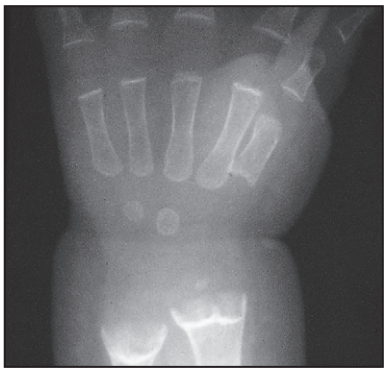


Figure 12.18 X-ray of the child's wrist showing rickets. The ends of the radius and ulna are expanded, rarefied and cup-shaped and the bones are poorly mineralised.

Management

Nutritional rickets is managed by advice about a balanced diet, correction of predisposing risk factors and by the daily administration of vitamin D_3 (cholecalciferol). If compliance is an issue, a single oral high dose of vitamin D_3 can be given, followed by the daily maintenance dose. Healing occurs in 2–4 weeks and can be monitored from the lowering of alkaline phosphatase, increasing vitamin D levels and healing on X-rays, but complete reversal of bony deformities may take years.

Summary

Rickets

- Nutritional – has re-emerged in the UK in Asian and black infants exclusively breast-fed into late infancy
- Diagnosis – serum calcium is low or normal, phosphorus low, plasma alkaline phosphatase greatly increased, 25-hydroxyvitamin D low and parathyroid hormone elevated
- X-ray features – cupping and fraying of the metaphyses and widened epiphyseal plate.

Vitamin A deficiency

In developed countries, vitamin A (retinol) deficiency is seen as a complication of fat malabsorption when supplementation has been inadequate. Clinical manifestations under these circumstances are rare, except for impaired adaptation to dark light. Vitamin A deficiency is the commonest cause of blindness in developing countries. It causes eye damage (xerophthalmia), which may progress from night blindness to corneal ulceration and scarring. It also results in increased susceptibility to infection, especially measles. Prevention in developing countries with high prevalence is by giving young children a dose of vitamin A; in some countries food is fortified. Supplementation is recommended for children with measles.

Obesity

Obesity is the most common nutritional disorder affecting children and adolescents in the developed world. Its importance is in its short- and long-term complications (Box 12.7) and that obese children tend to become obese adults.

Box 12.7 Complications of obesity

- Orthopaedic – slipped upper femoral epiphysis, tibia vara (bow legs), abnormal foot structure and function
- Idiopathic intracranial hypertension (headaches, blurred optic disc margins)
- Hypoventilation syndrome (daytime somnolence; sleep apnoea; snoring; hypercapnia; heart failure)
- Gallbladder disease
- Polycystic ovarian syndrome
- Type 2 diabetes mellitus
- Hypertension
- Abnormal blood lipids
- Other medical sequelae, e.g. asthma, changes in left ventricular mass, increased risk of certain malignancies (endometrial, breast and colonic carcinoma)
- Psychological sequelae – low self-esteem, teasing, depression.

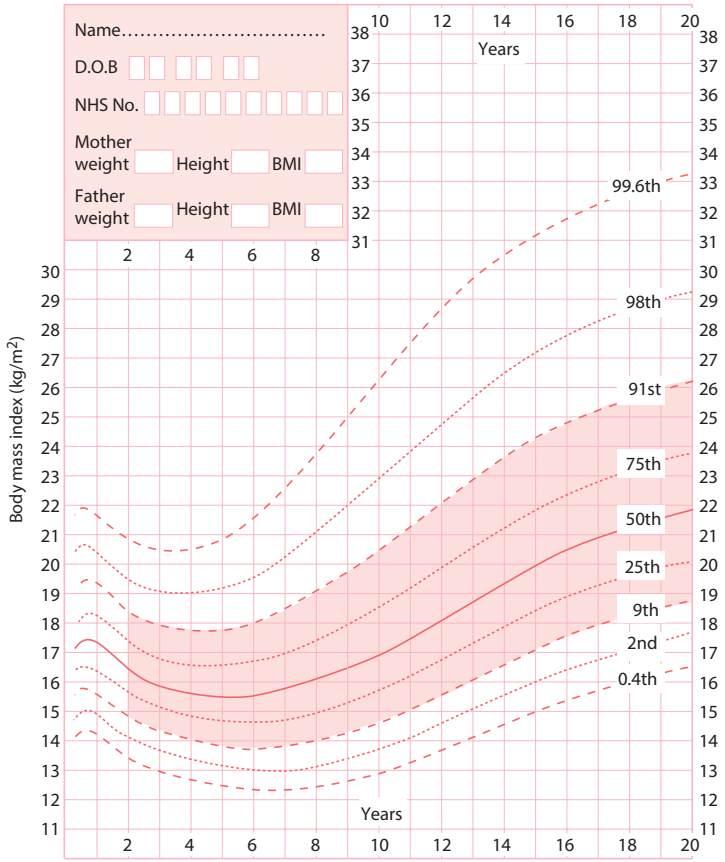


Figure 12.19 Body mass index (BMI) centile chart (Child Growth Foundation, reproduced with permission. Further supplies and information available at: http://www.healthforallchildren.co.uk. Accessed January 2011).

Definitions

In children, the body mass index (BMI = weight in kg/height in metres^2) is expressed as a BMI centile in relation to age and sex-matched population. By convention in the UK, the 1990 chart is used (Fig. 12.19). For clinical use, overweight is a BMI >91st centile, obese is a BMI >98th centile. Very severe obesity is >3.5 standard deviations above the mean; extreme obesity >4 standard deviations. For children over 12 years old, overweight is BMI ≥25, obese ≥30, very severe obesity BMI ≥35 and extreme obesity BMI ≥40 kg/m^2.

In 1995, 11% of males and 12.5% of females aged 2–15 years were obese. By 2006, these figures were 17.5% and 14.5%, respectively.

Aetiology

The reasons for this marked increase in prevalence are unclear but are due to changes in the environment and behaviour relating to diet and activity.

Energy-dense foods are now widely consumed, including high-fat fast foods and processed foods. However, there is no conclusive evidence that obese children eat more than children of normal weight. The National Food Survey showed that UK household energy intake has fallen since the 1970s, the amount of fruit purchased has increased by 75% and the intake of full fat milk decreased by 80%.

Children's energy expenditure has undoubtably decreased. Fewer children walk to school; transport in

cars has increased; less time at school is spent doing physical activities; and children spend more time in front of small screens (video-games, mobile phones, computers and television), rather than playing outside.

Children from low socioeconomic homes are more likely to be obese; females from the lowest socioeconomic quintile are 2.5 times more likely to be overweight when compared with the highest quintile.

Prevention

There are few randomised controlled trials and most involve complex packages of interventions. Interventions include decreased fat intake, increased fruit and vegetables, reduction in time spent in front of small screens, increased physical activity, and education. Of these, a reduction in time spent on small screens appears to be the most effective single factor.

Endogenous causes

Overnutrition accelerates linear growth and puberty. Obese children are therefore relatively tall and will usually be above the 50th centile for height. An endogenous cause, i.e. hypothyroidism and Cushing syndrome should be sought in short, obese children, in whom height velocity is decreased as height remains static in these conditions.

In children who are obese with learning disabilities, or who are dysmorphic, a syndrome should be considered. The commonest of these is Prader–Willi (obesity, hyperphagia, poor linear growth, dysmorphic facial features, hypotonia and undescended testes in males; see Chapter 8 and Figure 8.19. In severely obese children under the age of 3 years, gene defects, e.g. leptin deficiency, should be considered.

Management

Most obese children are managed in primary care. Specialist paediatric assessment is indicated in any child with complications (Box 12.7) or if an endogenous cause is suspected.

In the absence of evidence from randomised controlled trials, a pragmatic approach in any individual child based on consensus criteria has to be adopted (Box 12.8). Treatment should be considered where the child is above the 98th centile for BMI and the family are willing to make the necessary difficult lifestyle changes. Weight maintenance is a more realistic goal than weight reduction and will result in a demonstrable fall in BMI on centile chart as height increases. It can only be achieved by sustained changes in lifestyle:

- Healthier eating – no sugar-containing juices or fizzy drinks; decrease food portion size by 10–20%; increase protein- and non-carbohydrate-containing vegetables, discourage snacking and encourage family meals
- An increase in habitual physical activity to 60 min of moderate to vigorous daily physical activity
- Reduce physical inactivity (e.g. small screen time) during leisure time to less than an average of 2 h per day.

Box 12.8 Main messages for patients and parents about obesity

- Incidence of obesity in children is increasing
- Obesity is a health concern in itself and also increases the risk of other serious health problems, such as high blood pressure, diabetes and psychological distress
- An obese child tends to become an obese adult
- Obesity in children may be prevented and treated by increasing physical activity/decreasing physical inactivity (e.g. small screen time) and encouraging a well-balanced and healthy diet
- Lifestyle changes involve making small gradual changes to behaviour
- Family support is necessary for treatment to succeed
- Generally, the aim of treatment is to help children maintain their weight (so that they can 'grow into it')
- Most children are not obese because of an underlying medical problem but as a result of their lifestyle.

Adapted from Scottish Intercollegiate Guidelines Network, SIGN.

Drug treatment and surgery

Drug treatment has a part to play in children over the age of 12 who have extreme obesity (BMI>40 kg/m^2) or have a BMI>35 kg/m^2 and complications of obesity. It is recommended that drug treatment should only be considered after dietary, exercise and behavioural approaches have been started (NICE 2006).

Orlistat is a lipase inhibitor, which reduces the absorption of dietary fat and thus produces steatorrhoea. Fat intake should be reduced to avoid the unpleasant gastrointestinal side-effects. *Metformin* is a biguanide that increases insulin sensitivity, decreases gluconeogenesis and decreases gastrointestinal glucose absorption.

If there is evidence of insulin insensitivity (*Acanthosis nigricans*, see Fig. 25.2), metformin should be considered. Orlistat may be appropriate if fat intake is high.

Bariatric surgery is generally not considered appropriate in children or young people unless they have almost achieved maturity, have very severe or extreme obesity with complications, e.g. type 2 diabetes or hypertension, and all other interventions have failed to achieve or maintain weight loss. American data would suggest that laparoscopic adjustable gastric banding is the most appropriate operation.

Dental caries

Dental caries occurs as a result of exposure to organic acids produced by bacterial fermentation of carbohydrate, particularly sucrose (Fig. 12.20). Prevalence is now rising in young children. It is strongly related to socioeconomic deprivation.

Summary

Obesity

- An increasing major health issue for children, predisposing them to a wide range of medical and psychological problems in childhood and adult life, especially type 2 diabetes mellitus and cardiovascular disease
- Defined as a BMI >98th centile of the UK 1990 reference chart for age and sex; overweight is BMI >91st centile
- Exogenous causes (hypothyroidism and Cushing syndrome) of obesity are rare, and more likely in a child who is also short with falling height velocity; there are also some rare genetic syndromes
- Successful management requires sustained changes in lifestyle, with healthier eating, increased physical activity and reduction in physical inactivity
- Drug treatment and surgical intervention are only appropriate in a small number of children
- Lifestyle changes are difficult to achieve and even harder to maintain
- A cultural change in our society should be considered, e.g. removal of 'tuck shops' and vending machines with unhealthy food and drinks from schools.

Prevention involves:

- A reduction in plaque bacteria (brushing and flossing)
- Less frequent ingestion of carbohydrates
- Regular inspection by a dentist
- An optimal intake of fluoride up to puberty to improve the resistance of the tooth to damage; water fluoridation is one of the most effective ways of achieving this.

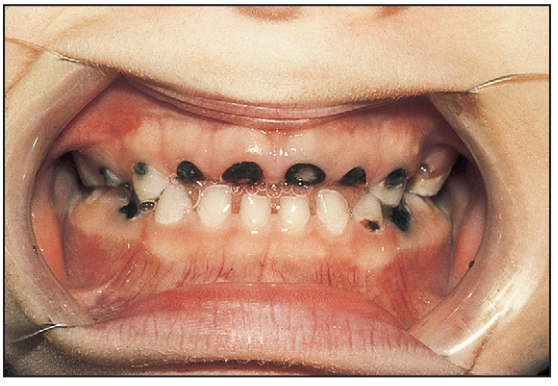


Figure 12.20 Dental caries. Prop-feeding infants when put to sleep with a bottle containing milk or other fermentable liquids places them at high risk of severe dental caries and should be discouraged.

Incorporation of fluoride in enamel by ionic substitution leads to replacement of calcium hydroxyapatite with calcium fluorapatite, which is less soluble in organic acids. In areas where drinking water contains a low concentration of fluoride, supplementation with fluoride drops or tablets is needed. Additionally, topical fluoride in toothpaste or mouthwashes is also advisable. Excess fluoride administration, before enamel has formed, may lead to mottled enamel (dental fluorosis).

Infants and children who are put to bed with a bottle containing fermentable liquid (milk or a sucrose-containing fruit juice) are at particular risk of developing severe dental caries. Characteristically, fluid collects around the upper anterior and posterior teeth, which become extensively damaged. Because of reduced salivation and swallowing during sleep, clearance and neutralisation of organic acids are also reduced. So-called 'prop feeding' should therefore be energetically discouraged. Infants fed on specialised formulae are also more at risk of developing dental caries because the carbohydrate in the milk is a glucose polymer.

Further reading

Beattie M, Dhawan A, Puntis J: *Paediatric Gastroenterology, Hepatology and Nutrition* (Oxford Specialist Handbooks in Paediatrics), Oxford, 2009, Oxford University Press .

Short handbook.

Websites (Accessed May 2011)

The Baby Friendly Initiative: Available at: http://www.babyfriendly.org.

Practical information on breast-feeding.

World Health Organization: Guidelines for malnutrition in developing countries. Available at: www.who.int.

Infant Feeding Survey: 2005. Available at: http://www.ic.nhs.uk/pubs/ifs2005.

NICE Guideline 43. Obesity: the prevention, identification, assessment and management of overweight and obesity in adults and children. 2006. Available at: http://guidance.nice.org.uk/CG43/Guidance/Section.

13

Gastroenterology

Vomiting	219	Malabsorption	233
Crying	222	Toddler diarrhoea	234
Acute abdominal pain	223	Inflammatory bowel disease	235
Recurrent abdominal pain	227	Constipation	237
Gastroenteritis	229		

Features of gastrointestinal disorders in children are:

- Vomiting, abdominal pain and diarrhoea are common and usually transient; serious causes are uncommon but important to identify
- Worldwide, gastroenteritis is responsible for 1.2 million deaths/year, one of the commonest causes of death in children <5 years old
- The number of children and adolescents developing inflammatory bowel disease is increasing
- In contrast to adults, bowel cancer is extremely rare.

Vomiting

Posseting and *regurgitation* are terms used to describe the non-forceful return of milk, but differ in degree. Posseting describes the small amounts of milk which often accompany the return of swallowed air ('wind'), whereas regurgitation describes larger, more frequent losses. Posseting occurs in nearly all babies from time to time, whereas regurgitation may indicate the presence of more significant gastro-oesophageal reflux.

Vomiting is the forceful ejection of gastric contents. It is a common problem in infancy and childhood (Fig. 13.1 and Box 13.1).

Box 13.1 'Red Flag' clinical features in the vomiting child

Bile-stained vomit	Intestinal obstruction
Haematemesis	Oesophagitis, peptic ulceration, oral/nasal bleeding
Projectile vomiting, in first few weeks of life	Pyloric stenosis
Vomiting at the end of paroxysmal coughing	Whooping cough (pertussis)
Abdominal tenderness/abdominal pain on movement	Surgical abdomen
Abdominal distension	Intestinal obstruction, including strangulated inguinal hernia
Hepatosplenomegaly	Chronic liver disease
Blood in the stool	Intussusception, gastroenteritis – salmonella or campylobacter
Severe dehydration, shock	Severe gastroenteritis, systemic infection (urinary tract infection, meningitis), diabetic ketoacidosis
Bulging fontanelle or seizures	Raised intracranial pressure
Failure to thrive	Gasto-oesophageal reflux, coeliac disease and other chronic gastrointestinal conditions

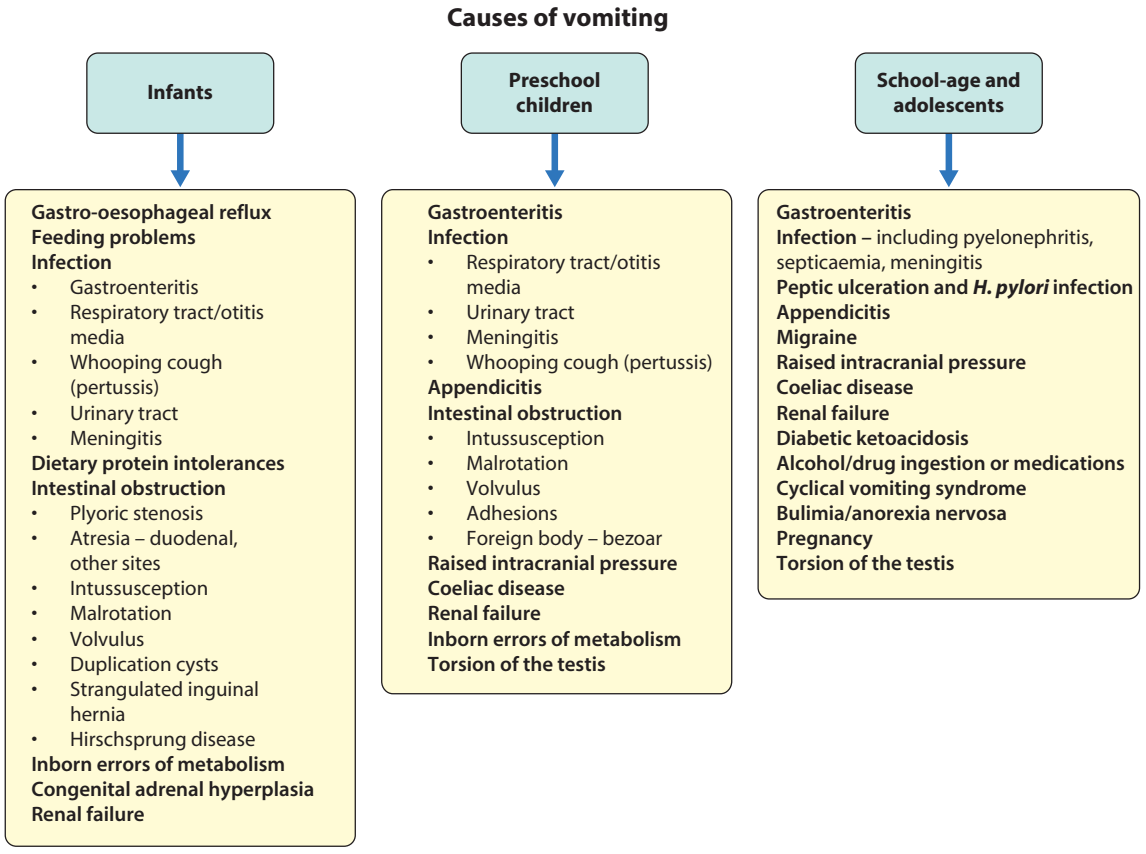


Figure 13.1 Causes of regurgitation/vomiting.

It is usually benign and is often caused by feeding disorders or mild gastro-oesophageal reflux or gastroenteritis. Potentially serious disorders need to be excluded if the vomiting is bilious or prolonged, or if the child is systemically unwell or failing to thrive. In infants, vomiting may be associated with infection outside the gastrointestinal tract, especially in the urinary tract and central nervous system. In intestinal obstruction, the more proximal the obstruction, the more prominent the vomiting and the sooner it becomes bile-stained (unless the obstruction is proximal to the ampulla of Vater). Intestinal obstruction is associated with abdominal distension, more marked in distal obstruction. 'Red Flag' clinical features suggesting significant organic pathology are listed in Box 13.1.

Summary

Vomiting in infants

- Common chronic causes are gastro-oesophageal reflux and feeding problems, e.g. force-feeding or overfeeding
- If transient, with other symptoms, e.g. fever, diarrhoea or runny nose and cough, most likely to be gastroenteritis or respiratory tract infection, but consider urine infection and meningitis
- If projectile at 2–7 weeks of age, exclude pyloric stenosis
- If bile stained, exclude intestinal obstruction, especially intussusception, malrotation and a strangulated inguinal hernia. Assess for dehydration and shock.

Gastro-oesophageal reflux

Gastro-oesophageal reflux is the involuntary passage of gastric contents into the oesophagus. It is extremely common in infancy. It is caused by inappropriate relaxation of the lower oesophageal sphincter as a result of functional immaturity. A predominantly fluid diet, a mainly horizontal posture and a short intra-abdominal length of oesophagus all contribute. While common in the first year of life, nearly all symptomatic reflux resolves spontaneously by 12 months of age. This is presumably due to a combination of maturation of the lower oesophageal sphincter, assumption of an upright posture and more solids in the diet.

Most infants with gastro-oesophageal reflux have recurrent regurgitation or vomiting but are putting on weight normally and are otherwise well, although the mess, smell and frequent changes of clothes is frustrating for carers.

Complications are listed in Box 13.2.

Severe reflux is more common in:

- children with cerebral palsy or other neurodevelopmental disorders, when energetic management, surgical if necessary, may transform the child's quality of life
- preterm infants, especially if coexistent bronchopulmonary dysplasia
- following surgery for oesophageal atresia or diaphragmatic hernia.

Box 13.2 Complications of gastro-oesophageal reflux

- Failure to thrive from severe vomiting
- Oesophagitis – haematemesis, discomfort on feeding or heartburn, iron deficiency anaemia
- Recurrent pulmonary aspiration – recurrent pneumonia, cough or wheeze, apnoea in preterm infants
- Dystonic neck posturing (Sandifer syndrome)
- Apparent life-threatening events (ALTE)

Investigation

Gastro-oesophageal reflux is usually diagnosed clinically and no investigations are required. However, they may be indicated if the history is atypical, complications are present or there is failure to respond to treatment. Investigations include:

- 24-hour oesophageal pH monitoring to quantify the degree of acid reflux (see Case History 13.1).
- 24-hour impedance monitoring. Available in some centres. Weakly acidic or non-acid reflux, which may cause disease, is also measured.
- Endoscopy with oesophageal biopsies to identify oesophagitis and exclude other causes of vomiting.

Case History

13.1 Severe gastro-oesophageal reflux

This infant (Fig. 13.2a) had a history of frequent regurgitation from the first few days of life. He developed two chest infections. Some of the vomits contained altered blood. A 24-hour oesophageal pH study showed severe gastro-oesophageal reflux (Fig. 13.2b,c). Endoscopy showed oesophagitis. He had probably had episodes of aspiration pneumonia. Symptoms resolved on treatment with feed thickeners and omeprazole. His parents also commented on how much better he slept at night. Treatment was reduced from 14 months of age and the symptoms did not recur.



Figure 13.2a A pH sensor has been placed in the lower oesophagus.

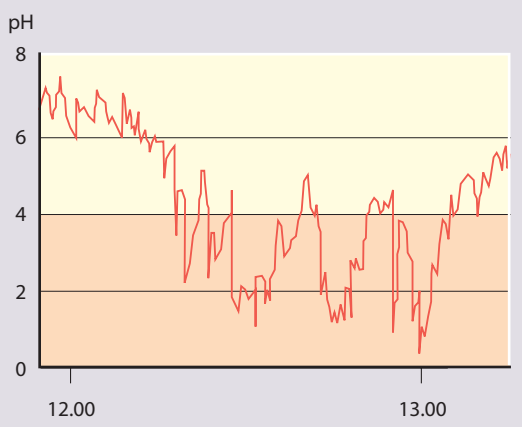


Figure 13.2b A section of the 24-hour oesophageal pH study showing severe reflux, with frequent drops in pH below 4.

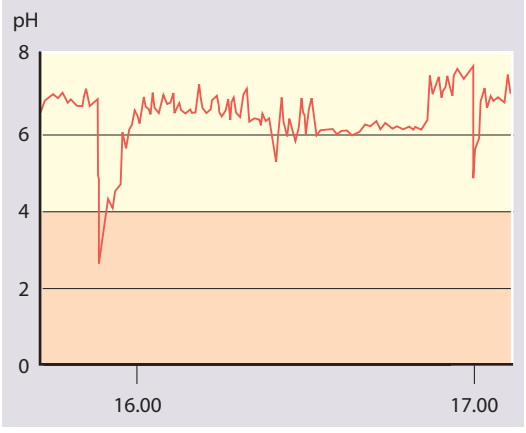


Figure 13.2c A section of a normal oesophageal pH study. The lower oesophageal pH is above 4 for most of the time.

Contrast studies of the upper gastrointestinal tract may support the diagnosis but are neither sensitive nor specific. They may be required to exclude underlying anatomical abnormalities in the oesophagus, stomach and duodenum, and to identify malrotation.

Management

Uncomplicated gastro-oesophageal reflux has an excellent prognosis and can be managed by parental reassurance, adding inert thickening agents to feeds (e.g. Nestargel, Carobel) and positioning in a 30° head-up prone position after feeds.

More significant gastro-oesophageal reflux disease is managed with acid suppression with either H_2 receptor antagonists (e.g. ranitidine) or proton pump inhibitors (e.g. omeprazole). These drugs reduce the volume of gastric contents and treat acid-related oesophagitis. Although the evidence for their use is poor, drugs which enhance gastric emptying (e.g. domperidone) may be tried. If the child fails to respond to these measures, other diagnoses such as cow's milk protein allergy should be considered and further investigations performed.

Surgical management is reserved for children with complications unresponsive to intensive medical treatment or oesophageal stricture. A Nissen fundoplication, in which the fundus of the stomach is wrapped around the intra-abdominal oesophagus, is performed either as an abdominal or laparoscopic procedure.

Summary

Gastro-oesophageal reflux

- Occurs in otherwise normal infants, but risk is increased if neuromuscular problems or surgery to the oesophagus or diaphragm
- Is treated if troublesome with upright positioning, feed thickening, medication and sometimes fundoplication
- Investigations are performed if diagnosis is unclear or complications occur.

Pyloric stenosis

In pyloric stenosis, there is hypertrophy of the pyloric muscle causing gastric outlet obstruction. It presents at between 2 and 7 weeks of age, irrespective of gestational age. It is more common in boys (4:1), particularly first-borns, and there may be a family history, especially on the maternal side.

Clinical features are:

- Vomiting, which increases in frequency and forcefulness over time, ultimately becoming projectile
- Hunger after vomiting until dehydration leads to loss of interest in feeding
- Weight loss if presentation is delayed.

A hypochloraemic metabolic alkalosis with a low plasma sodium and potassium occurs as a result of vomiting stomach contents.

Diagnosis

Unless immediate fluid resuscitation is required, a test feed is performed. The baby is given a milk feed, which will calm the hungry infant, allowing examination. Gastric peristalsis may be seen as a wave moving from left to right across the abdomen (Fig. 13.3a). The pyloric mass, which feels like an olive, is usually palpable in the right upper quadrant (Fig. 13.3b). If the stomach is overdistended with air, it will need to be emptied by a nasogastric tube to allow palpation. Ultrasound examination is helpful (Fig. 13.3c) if the diagnosis is in doubt.

Management

The initial priority is to correct any fluid and electrolyte disturbance with intravenous fluids (0.45% saline and 5% dextrose with potassium supplements). Once hydration and acid–base and electrolytes are normal, definitive treatment by pyloromyotomy can be performed. This involves division of the hypertrophied muscle down to, but not including, the mucosa (Fig. 13.3d). The operation can be performed either as an open procedure via a periumbilical incision or laparoscopically. Postoperatively, the child can usually be fed within 6 h and discharged within 2 days of surgery.

Summary

Pyloric stenosis

- More common in boys and those with a maternal family history
- Signs are: visible gastric peristalsis, palpable abdominal mass on test feed and possible dehydration
- Associated with hyponatraemia, hypokalaemia and hypochloraemic alkalosis
- Diagnosis may be confirmed by ultrasound
- Treated by surgery after rehydration and correction of electrolyte imbalance.

Crying

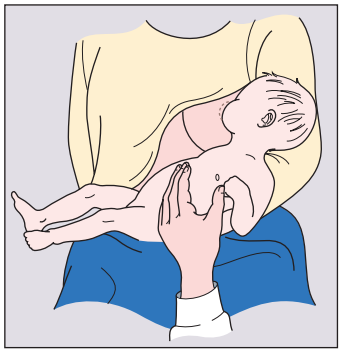
The time healthy babies cry for is highly variable. In most, it represents the baby's response to hunger and discomfort. Reassurance and advice on appropriate feeding, wrapping and care will usually suffice.

Some babies cry for prolonged periods in spite of feeding and comforting and this is distressing for all concerned. It can engender a feeling of anxiety, helplessness and depression in the carer, particularly if they are inexperienced or poorly supported. It has also been suggested that the emotional climate within a home may be transmitted to a baby, and that in some

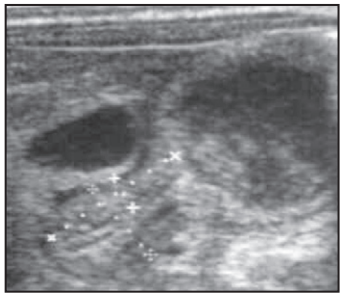
Pyloric stenosis



(a)



(b)



(c)



(d)

Figure 13.3 **(a)** Visible gastric peristalsis in an infant with pyloric stenosis. **(b)** Diagram showing a test feed being performed to diagnose pyloric stenosis. The pyloric mass feels like an 'olive' on gentle, deep palpation halfway between the midpoint of the anterior margin of the right ribcage and the umbilicus. **(c)** Ultrasound examination showing pyloric stenosis. **(d)** Pyloric stenosis at operation showing pale, thick pyloric muscle and pyloromyotomy incision.

instances, tense, anxious or irritable caregivers are more likely to have fretful babies. The complaint that a baby is 'always crying' may also be a pointer to potential or actual non-accidental injury.

A significant cause for the crying is identified in a minority of infants. If of sudden onset, it may be due to a urinary tract, middle ear or meningeal infection, to pain from an unrecognised fracture, oesophagitis or torsion of the testis. Severe nappy rash, constipation or coeliac disease may produce a miserable, crying infant. Preterm infants who have spent several weeks in hospital can be difficult to settle, as can infants with a chronic neurological disorder, e.g. cerebral palsy. On the basis of countless reports of parents, eruption of teeth is painful in some infants. However, teething does not cause vomiting, diarrhoea, high fever or seizures.

Infant 'colic'

The term 'colic' is used to describe a common symptom complex which occurs during the first few months of life. Paroxysmal, inconsolable crying or screaming often accompanied by drawing up of the knees and passage of excessive flatus takes place several times a day, particularly in the evening. There is no firm evidence that the cause is gastrointestinal, but this is often suspected. The condition occurs in up to 40% of babies. It typically occurs in the first few weeks of life and resolves by 4 months of age. The condition is benign but it is very frustrating and worrying for parents and may precipitate non-accidental injury in infants already at risk. Support and reassurance should be given. Gripe water is often recommended but is of unproven benefit. If severe and persistent, it may be due to a cow's milk protein allergy or gastro-oesophageal reflux and an empirical 2-week trial of a whey hydrolysate formula followed by a trial of anti-reflux treatment may be considered.

Acute abdominal pain

Assessment of the child with acute abdominal pain requires considerable skill. The differential diagnosis of acute abdominal pain in children is extremely wide, encompassing non-specific abdominal pain, surgical causes and medical conditions (Fig. 13.4). In nearly half of the children admitted to hospital, the pain resolves undiagnosed. In young children it is essential not to delay the diagnosis and treatment of acute appendicitis, as progression to perforation can be rapid. It is easy to belittle the clinical signs of abdominal tenderness in young children. Of the surgical causes, appendicitis is by far the most common. The testes, hernial orifices

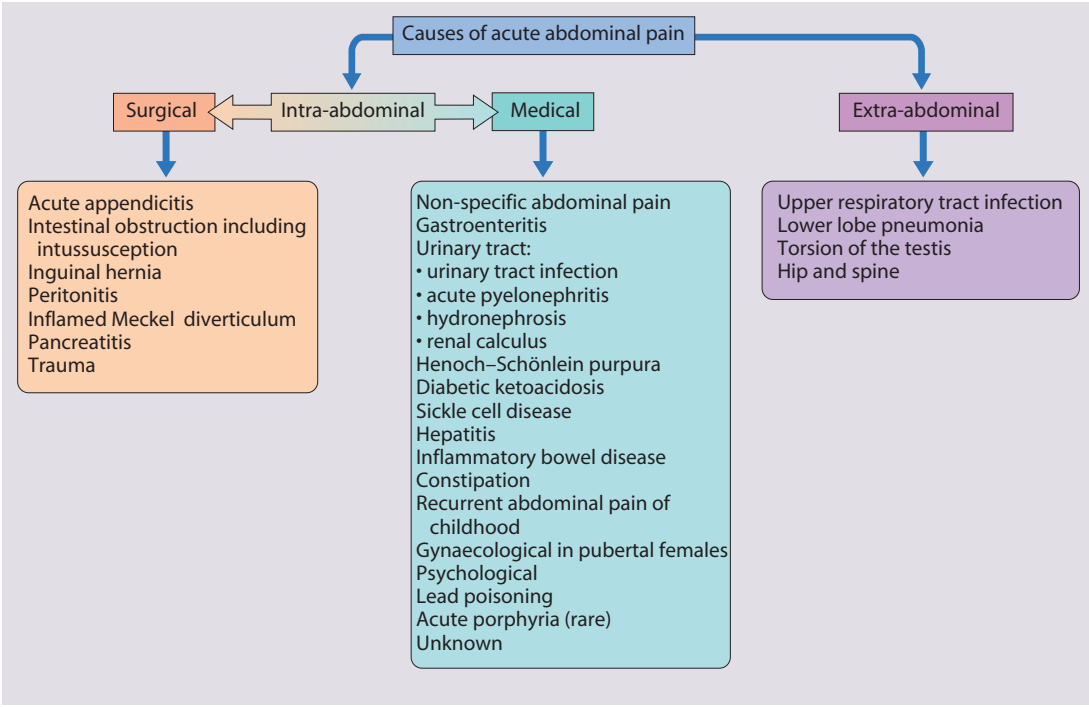


Figure 13.4 Causes of acute abdominal pain.

and hip joints must always be checked. It is noteworthy that:

- Lower lobe pneumonia may cause pain referred to the abdomen
- Primary peritonitis is seen in patients with ascites from nephrotic syndrome or liver disease
- Diabetic ketoacidosis may cause severe abdominal pain
- Urinary tract infection, including acute pyelonephritis, is a relatively uncommon cause of acute abdominal pain, but must not be missed. It is important to test a urine sample, in order to identify not only diabetes mellitus but also conditions affecting the liver and urinary tract.

Acute appendicitis

Acute appendicitis is the commonest cause of abdominal pain in childhood requiring surgical intervention (Fig. 13.5). Although it may occur at any age, it is very uncommon in children <3 years old. The clinical features of acute uncomplicated appendicitis are:

- Symptoms
  - Anorexia
  - Vomiting (usually only a few times)
  - Abdominal pain, initially central and colicky (appendicular midgut colic), but then localising to the right iliac fossa (from localised peritoneal inflammation)
- Signs
  - Flushed face with oral fetor
  - Low-grade fever 37.2–38°C
  - Abdominal pain aggravated by movement, e.g. on walking, coughing, jumping, bumps on the road during a car journey
  - Persistent tenderness with guarding in the right iliac fossa (McBurney's point).

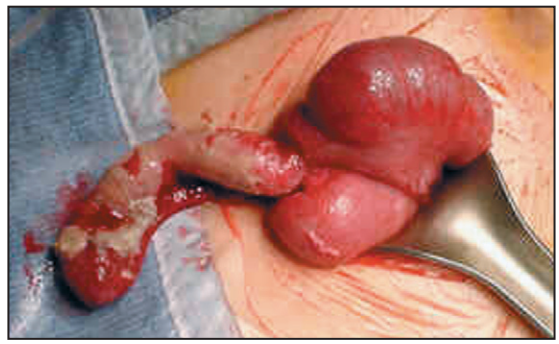


Figure 13.5 Appendicitis at operation showing a perforated acutely inflamed appendix covered in fibrin.

In preschool children:

- The diagnosis is more difficult, particularly early in the disease
- Faecoliths are more common and can be seen on a plain abdominal X-ray
- Perforation may be rapid, as the omentum is less well developed and fails to surround the appendix, and the signs are easy to underestimate at this age.

With a retrocaecal appendix, localised guarding may be absent, and in a pelvic appendix there may be few abdominal signs.

Appendicitis is a progressive condition and so repeated observation and clinical review every few hours are key to making the correct diagnosis, avoiding delay on the one hand and unnecessary laparotomy on the other.

No laboratory investigation or imaging is consistently helpful in making the diagnosis. A neutrophilia is not always present on a full blood count. White blood cells or organisms in the urine are not uncommon in appendicitis as the inflamed appendix may be adjacent to the ureter or bladder. Although ultrasound is no substitute for regular clinical review, it may support the clinical diagnosis (thickened, non-compressible appendix with increased blood flow); demonstrate associated complications such as abscess, perforation or appendix mass; and exclude other pathology causing the symptoms. In some centres, laparoscopy is available to see whether or not the appendix is inflamed.

Appendicectomy is straightforward in uncomplicated appendicitis. Complicated appendicitis includes the presence of an appendix mass, an abscess or perforation. If there is generalised guarding consistent with perforation, fluid resuscitation and intravenous antibiotics are given prior to laparotomy. If there is a palpable mass in the right iliac fossa and there are no signs of generalised peritonitis, it may be reasonable to elect for conservative management with intravenous antibiotics, with appendicectomy being performed after several weeks. If symptoms progress, laparotomy is indicated.

Non-specific abdominal pain and mesenteric adenitis

Non-specific abdominal pain (NSAP) is abdominal pain which resolves in 24–48 h. The pain is less severe than in appendicitis, and tenderness in the right iliac fossa is variable. It is often accompanied by an upper respiratory tract infection with cervical lymphadenopathy. In some of these children, the abdominal signs do not resolve and an appendicectomy is performed. The diagnosis of mesenteric adenitis can only be made definitively in those children in whom large mesenteric nodes are seen at laparotomy or laparoscopy and whose appendix is normal.

Intussusception

Intussusception describes the invagination of proximal bowel into a distal segment. It most commonly involves ileum passing into the caecum through the ileocaecal valve (Fig. 13.6a). Intussusception is the commonest cause of intestinal obstruction in infants after the neonatal period. Although it may occur at any age, the peak age of presentation is between 3 months and 2 years. The most serious complication is stretching and constriction of the mesentery resulting in venous obstruction, causing engorgement and bleeding from the bowel mucosa, fluid loss and subsequently bowel perforation, peritonitis and gut necrosis. Prompt diagnosis, immediate fluid resuscitation and urgent reduction of the intussusception are essential to avoid complications.

Presentation is typically with:

- Paroxysmal, severe colicky pain and pallor – during episodes of pain, the child becomes pale, especially around the mouth, and draws up his legs. He initially recovers between painful episodes, but subsequently becomes increasingly lethargic
- May refuse feeds, may vomit, which may become bile-stained depending on the site of the intussusception
- A sausage-shaped mass – often palpable in the abdomen (Fig. 13.6b)
- Passage of a characteristic redcurrant jelly stool comprising blood-stained mucus – this is a characteristic sign but tends to occur later in the illness and may be first seen after a rectal examination
- Abdominal distension and shock.

Usually, no underlying intestinal cause for the intussusception is found, although there is some evidence that viral infection leading to enlargement of Peyer's patches may form the lead point of the intussusception. An identifiable lead point such as a Meckel diverticulum or polyp is more likely to be present in children over 2 years old. Intravenous fluid resuscitation is likely to be required immediately, as there is often pooling of fluid in the gut, which may lead to hypovolaemic shock.

An X-ray of the abdomen may show distended small bowel and absence of gas in the distal colon or rectum. Sometimes the outline of the intussusception itself can be visualised. Abdominal ultrasound is helpful both to confirm the diagnosis and to check response to treatment. Unless there are signs of peritonitis, reduction of the intussusception by rectal air insufflation is usually attempted by a radiologist (Fig. 13.6c). This procedure should only be carried out once the child has been

Summary

Acute abdominal pain in older children and adolescents

- Exclude medical causes, in particular lower lobe pneumonia, diabetic ketoacidosis, hepatitis, pyelonephritis
- Check for strangulated inguinal hernia or torsion of the testis in boys
- On palpating the abdomen in children with acute appendicitis, guarding and rebound tenderness are often absent or unimpressive, but pain from peritoneal inflammation may be demonstrated on coughing, walking or jumping
- To distinguish between acute appendicitis and non-specific abdominal pain may require close monitoring and repeated evaluation in hospital.

Intussusception

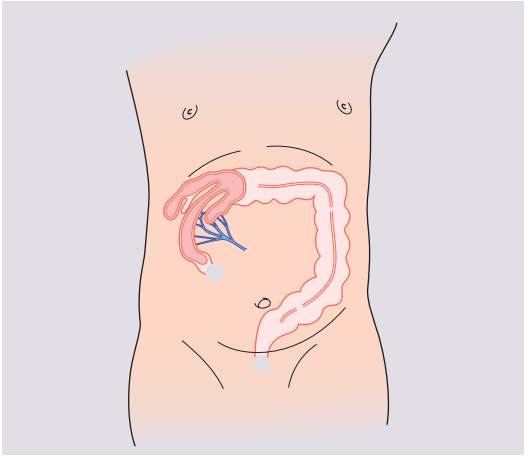


Figure 13.6a Intussusception, showing why the blood supply to the gut rapidly becomes compromised, making relief of this form of obstruction urgent.



Figure 13.6b A child with an intussusception. The mass can be seen in the upper abdomen. The child has become shocked.

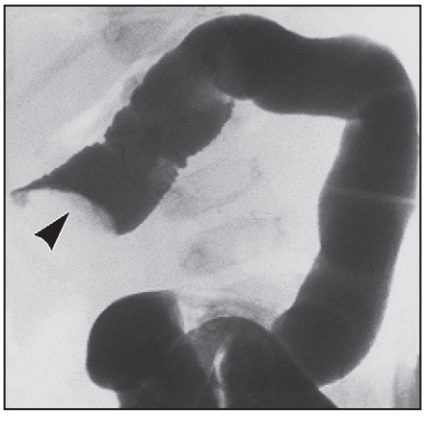


Figure 13.6c An abdominal X-ray demonstrating an intussusception (see arrowhead), taken during reduction by air insufflated per rectum.

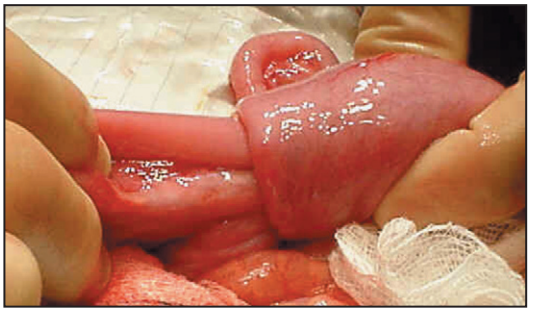


Figure 13.6d Intussusception at operation showing the ileum entering the caecum. The surgeon is squeezing the colon to reduce the intussusception.

Summary

Intussusception

- Usually occurs between 3 months and 2 years of age
- Clinical features are paroxysmal, colicky pain with pallor, abdominal mass, redcurrant jelly stool
- Shock is an important complication and requires urgent treatment
- Reduction is attempted by rectal air insufflation unless peritonitis is present
- Surgery is required if reduction with air is unsuccessful or for peritonitis.

resuscitated and is under the supervision of a paediatric surgeon in case the procedure is unsuccessful or bowel perforation occurs. The success rate of this procedure is about 75%. The remaining 25% require operative reduction (Fig. 13.6d). Recurrence of the intussusception occurs in less than 5% but is more frequent after hydrostatic reduction.

Meckel diverticulum

Around 2% of individuals have an ileal remnant of the vitello-intestinal duct, a Meckel diverticulum, which contains ectopic gastric mucosa or pancreatic tissue. Most are asymptomatic but they may present with severe rectal bleeding, which is classically neither bright red nor true melaena. Other forms of

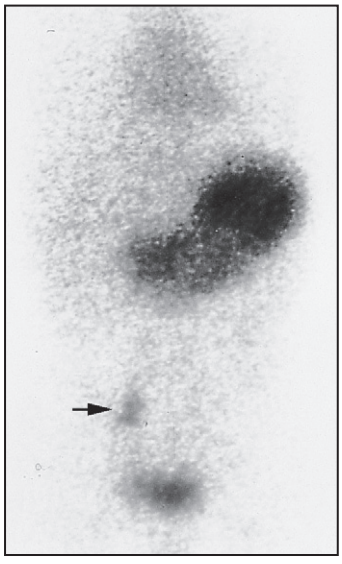


Figure 13.7 Technetium scan showing uptake by ectopic gastric mucosa in a Meckel diverticulum in the right iliac fossa.

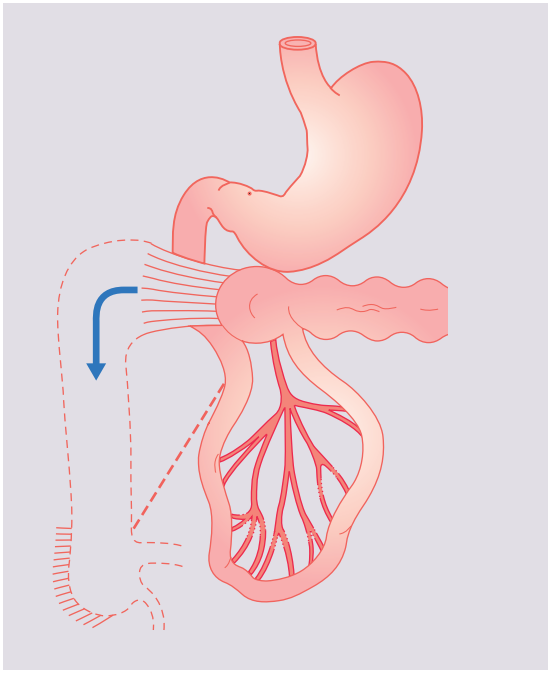


Figure 13.8 The commonest form of malrotation, with the caecum remaining high and fixed to the posterior abdominal wall. There are Ladd bands obstructing the duodenum. Dotted lines show normal anatomy.

presentation include intussusception, volvulus around a band, or diverticulitis which mimics appendicitis. A technetium scan will demonstrate increased uptake by ectopic gastric mucosa in 70% of cases (Fig. 13.7). Treatment is by surgical resection.

Summary

Meckel diverticulum

- Occurs in 2% of individuals.
- Generally asymptomatic, but may present with bleeding (which may be life-threatening), intussusception, volvulus or diverticulitis.
- Treatment is by surgical resection.

Malrotation

During rotation of the small bowel in fetal life, if the mesentery is not fixed at the duodenojejunal flexure or in the ileocaecal region, its base is shorter than normal, and is predisposed to volvulus. Ladd bands may cross the duodenum, contributing to bowel obstruction (Fig. 13.8).

There are two presentations:

- Obstruction
- Obstruction with a compromised blood supply.

Obstruction with bilious vomiting is the usual presentation in the first few days of life but can be seen at a later age. Any child with dark green vomiting needs an urgent upper gastrointestinal contrast study to assess intestinal rotation, unless signs of vascular compromise are present, when an urgent laparotomy is needed.

At operation, the volvulus is untwisted, the duodenum mobilised and the bowel placed in the non-rotated position with the duodenojejunal flexure on the right and the caecum and appendix on the left. The malrotation is not 'corrected', but the mesentery broadened. The appendix is generally removed to avoid diagnostic confusion in the event the child subsequently has symptoms suggestive of appendicitis.

Summary

Malrotation

- Uncommon but important to diagnose
- Usually presents in the first 1–3 days of life with intestinal obstruction from Ladd bands obstructing the duodenum or volvulus
- May present at any age with volvulus causing obstruction and ischaemic bowel
- Clinical features are bilious vomiting, abdominal pain and tenderness from peritonitis or ischaemic bowel
- An urgent upper gastrointestinal contrast study is indicated if there is bilious vomiting
- Treatment is urgent surgical correction.

Recurrent abdominal pain

Recurrent abdominal pain (RAP) is a common childhood problem. It is often defined as pain sufficient to interrupt normal activities and lasts for at least

3 months. It occurs in about 10% of school-age children. A cause (see Summary box) is identified in <10%. The pain is characteristically periumbilical and the children are otherwise entirely well. The widely held belief that they have psychogenic pain is without foundation, a number of studies having failed to show a difference between such children and their families and controls. In some children, it may however be a manifestation of stress (see Chapter 23) or it may become part of a vicious cycle of anxiety with escalating pain leading to family distress and demands for increasingly invasive investigations. There is evidence that anxiety may lead to altered bowel motility, which may be perceived by the child as pain.

It is increasingly recognised that many will have one of three distinct symptom constellations resulting from functional abnormalities of gut motility or enteral neurons – irritable bowel syndrome (most common), abdominal migraine or functional dyspepsia.

Management

The aim is to identify any serious cause without subjecting the child to unnecessary investigation, while providing reassurance to the child and parents. To do this, a full history and thorough examination is required, which includes inspection of the perineum for anal fissures. The child's growth should be checked.

A urine microscopy and culture is mandatory as urinary tract infections may cause pain in the absence of other symptoms or signs. An abdominal ultrasound is particularly helpful in excluding gall stones and pelviureteric junction (PUJ) obstruction.

Although there are many potential organic causes, most are rare and further investigations should be performed only if clinically indicated.

With irritable bowel syndrome and functional dyspepsia, it can be helpful to explain to both the child and parents that 'sometimes the insides of the intestine become so sensitive that some children can feel the food going round the bends'. It is also necessary to make a distinction between 'serious' and 'dangerous'. These disorders can be serious, if, for example, they lead to substantial loss of schooling, but they are not dangerous.

The long-term prognosis is that:

- about half of affected children rapidly become free of symptoms
- in one-quarter, the symptoms take some months to resolve
- in one-quarter, symptoms continue or return in adulthood as migraine, irritable bowel syndrome, or functional dyspepsia.

Abdominal migraine

Abdominal migraine is often associated with abdominal pain in addition to headaches, and in some children the abdominal pain predominates. The attacks of abdominal pain are midline associated with vomiting and facial pallor. There is usually a personal or family history of migraine.

Irritable bowel syndrome

This disorder, also common in adults, is associated with altered gastrointestinal motility and an abnormal sensation of intra-abdominal events. Symptoms may occur following a gastrointestinal infection. Studies of pressure changes within the small intestine of children with irritable bowel syndrome suggest that abnormally forceful contractions occur. It has also been shown that affected adults experience pain on inflation of balloons in the intestine at substantially lower volumes than do controls. There is therefore an interplay between these two factors, both of which are modulated by psychosocial factors such as stress and anxiety.

There is often a positive family history and a characteristic set of symptoms, although not all patients experience every symptom:

- Abdominal pain, often worse before or relieved by defecation
- Explosive, loose or mucousy stools
- Bloating
- Feeling of incomplete defecation
- Constipation (often alternating with normal or loose stools).

Peptic ulceration, gastritis and functional dyspepsia

The greater use of endoscopy in children and the identification of the Gram-negative organism *Helicobacter pylori* (*H. pylori*) in association with antral gastritis have focused attention on it as a potential cause of abdominal pain in children. In adults, there is substantial evidence that *H. pylori* is a strong predisposing factor to duodenal ulcers. This association in children is much less clear. Duodenal ulcers are uncommon in children but should be considered in those with epigastric pain, particularly if it wakes them at night, or when there is a history of peptic ulceration in a first-degree relative.

H. pylori causes a nodular antral gastritis which may be associated with abdominal pain and nausea. It is usually identified in gastric antral biopsies. The organism produces urease, which forms the basis for a laboratory test on biopsies, and the ^{13}C breath test following the administration of ^{13}C-labelled urea by mouth. Stool antigen for *H. pylori* may be positive in infected children. Serological tests are unreliable in children.

Children in whom peptic ulceration is suspected should be treated with proton pump inhibitors, e.g. omeprazole, and if investigations suggest they have an *H. pylori* infection, eradication therapy should be given (amoxicillin and metronidazole or clarithromycin). Those that fail to respond to treatment or whose symptoms recur on stopping treatment should have an upper GI endoscopy and, if this is normal, functional dyspepsia is diagnosed.

As well as having symptoms of peptic ulceration, children with functional dyspepsia have rather more non-specific symptoms, including early satiety, bloating and postprandial vomiting and may have delayed gastric emptying as a result of gastric dysmotility. Treatment is difficult but some children respond to a hypoallergenic diet.

Summary

Causes and assessment of the child with recurrent abdominal pain

>90% no structural cause identified

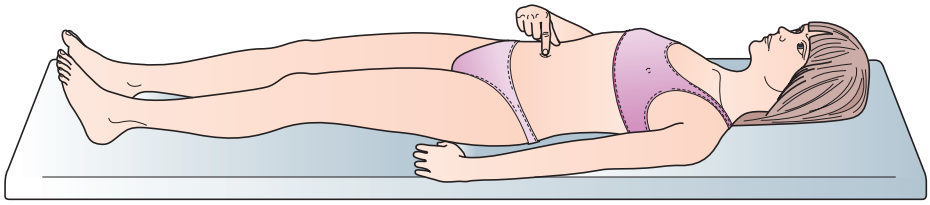
Gastrointestinal
- Irritable bowel syndrome
- Constipation
- Non-ulcer dyspepsia
- Abdominal migraine
- Gastritis and peptic ulceration
- Inflammatory bowel disease
- Malrotation

Gynaecological
- Dysmenorrhoea
- Ovarian cysts
- Pelvic inflammatory disease

Psychosocial – bullying, abuse, stress, etc. – a small proportion

Hepatobility/pancreatic
- Hepatitis
- Gall stones
- Pancreatitis

Urinary tract
- Urinary tract infection
- Pelvi-ureteric junction (PUJ) obstruction



Symptoms and signs that suggest organic disease:
- Epigastric pain at night, haematemesis (duodenal ulcer)
- Diarrhoea, weight loss, growth failure, blood in stools (inflammatory bowel disease)
- Vomiting (pancreatitis)
- Jaundice (liver disease)
- Dysuria, secondary enuresis (urinary tract infection)
- Bilious vomiting and abdominal distension (malrotation)

Gastroenteritis

In developing countries, gastroenteritis remains a major cause of child mortality. In developed countries, it is a cause of significant morbidity, particularly in younger children. In the UK, approximately 10% of under 5 year olds annually present to health services with gastroenteritis and it remains a common reason for hospital admission in young children.

The most frequent cause of gastroenteritis in developed countries is rotavirus infection, which accounts for up to 60% of cases in children <2 years of age, particularly during the winter and early spring. An effective vaccine against rotavirus is now available, but has not been adopted into the national immunisation programme. Other viruses, particularly adenovirus, norovirus, calicivirus, coronavirus and astrovirus, may cause outbreaks.

Bacterial causes are less common in developed countries and are suggested by the presence of blood in the stools. *Campylobacter jejuni* infection, the commonest of the bacterial infections in developed countries, is often associated with severe abdominal pain. *Shigella* and some salmonellae produce a dysenteric type of infection, with blood and pus in the stool, pain and tenesmus. *Shigella* may be accompanied by high fever. Cholera and enterotoxigenic *E. coli* infection are associated with profuse, rapidly dehydrating diarrhoea. However, clinical features act as a poor guide to the pathogen.

Box 13.3 Conditions which can mimic gastroenteritis

Systemic infection	Septicaemia, meningitis
Local infections	Respiratory tract infection, otitis media, hepatitis A, urinary tract infection
Surgical disorders	Pyloric stenosis, intussusception, acute appendicitis, necrotising enterocolitis, Hirschsprung disease
Metabolic disorder	Diabetic ketoacidosis
Renal disorder	Haemolytic uraemic syndrome
Other	Coeliac disease, cow's milk protein intolerance, adrenal insufficiency

In gastroenteritis there is a sudden change to loose or watery stools often accompanied by vomiting. There may be contact with a person with diarrhoea and/or vomiting or recent travel abroad. A number of disorders may masquerade as gastroenteritis (Box 13.3) and,

when in doubt, hospital referral is essential. Dehydration leading to shock is the most serious complication and its prevention or correction is the main aim of treatment.

The following children are at increased risk of dehydration:

- Infants, particularly those under 6 months of age or those born with low birthweight.
- If they have passed ≥6 diarrhoeal stools in the previous 24 h
- If they have vomited three or more times in the previous 24 h
- If they have been unable to tolerate (or not been offered) extra fluids
- If they have malnutrition.

Infants are at particular risk of dehydration because they have a greater surface area to weight ratio than older children, leading to greater insensible water losses (300 ml/m^2 per day, equivalent in infants to 15–17 ml/kg per day). They have higher basal fluid requirements (100–120 ml/kg per day, i.e. 10–12% of bodyweight) and immature renal tubular reabsorption. In addition, they are unable to obtain fluids for themselves when thirsty.

Assessment

Clinical assessment of dehydration is important but difficult. The most accurate measure of dehydration is the degree of weight loss during the diarrhoeal illness. A recent weight measurement is useful but is often not available and may be misleading if the child had clothes on or the different measuring scales are not accurate. The history and examination are used to assess the degree of dehydration as:

- No clinically detectable dehydration (usually <5% loss of body weight)
- Clinical dehydration (usually 5–10%)
- Shock (usually >10%) (Fig. 13.9 and Table 13.1). Shock must be identified without delay.

Isonatraemic and hyponatraemic dehydration

In dehydration, there is a total body deficit of sodium and water. In most instances, the losses of sodium and water are proportional and plasma sodium remains within the normal range (isonatraemic dehydration). When children with diarrhoea drink large quantities of water or other hypotonic solutions, there is a greater net loss of sodium than water, leading to a fall in plasma sodium (hyponatraemic dehydration). This leads to a shift of water from extra- to intracellular compartments. The increase in intracellular volume leads to an increase in brain volume, which may result in convulsions, whereas the marked extracellular depletion leads to a greater degree of shock per unit of water loss. This form of dehydration is more common in poorly nourished infants in developing countries.

Hypernatraemic dehydration

Infrequently, water loss exceeds the relative sodium loss and plasma sodium concentration increases (hypernatraemic dehydration). This usually results from high insensible water losses (high fever or hot, dry environment) or from profuse, low-sodium diarrhoea. The extracellular fluid becomes hypertonic with respect to the intracellular fluid, which leads to a shift of water into the extracellular space from the intracellular compartment. Signs of extracellular fluid depletion are therefore less per unit of fluid loss, and depression of the fontanelle, reduced tissue elasticity and sunken eyes are less obvious. This makes this form of dehydration more difficult to recognise clinically, particularly in an obese infant. It is a particularly dangerous form of dehydration as water is drawn out of the brain and cerebral shrinkage within a rigid skull may lead to jittery movements, increased muscle tone

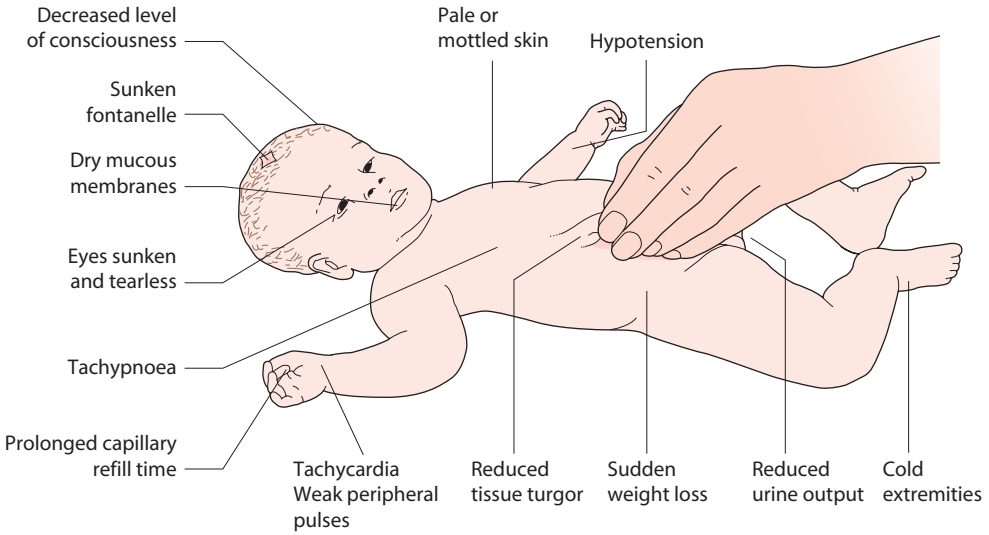


Figure 13.9 Clinical features of shock from dehydration in an infant.

Table 13.1 Clinical assessment of dehydration

	No clinical dehydration	Clinical dehydration	Shock
General appearance	Appears well	Appears unwell or deteriorating 🚩	Appears unwell or deteriorating
Conscious level	Alert and responsive	Altered responsiveness, e.g. irritable, lethargic 🚩	Decreased level of consciousness
Urine output	Normal	Decreased	Decreased
Skin colour	Normal	Normal	Pale or mottled
Extremities	Warm	Warm	Cold
Eyes	Normal	Sunken 🚩	Grossly sunken
Mucous membranes	Moist	Dry	Dry
Heart rate	Normal	Tachycardia 🚩	Tachycardia
Breathing	Normal	Tachypnoea 🚩	Tachypnoea
Peripheral pulses	Normal	Normal	Weak
Capillary refill time	Normal	Normal	Prolonged (>2 s)
Skin turgor	Normal	Reduced 🚩	Reduced
Blood pressure	Normal	Normal	Hypotension (indicates decompensated shock)

🚩; 'Red Flag' sign – helps to identify children at risk of progression to shock.
The more numerous and more pronounced the symptoms and signs, the greater the severity of dehydration.
From NICE Guideline, Diarrhoea and vomiting in children, 2009.

with hyperreflexia, altered consciousness, seizures and multiple, small cerebral haemorrhages. Transient hyperglycaemia occurs in some patients with hypernatraemic dehydration; it is self-correcting and does not require insulin.

Investigation

Usually, no investigations are indicated. Stool culture is required if the child appears septic, if there is blood or mucus in the stools or the child is immunocompromised. It may be indicated following recent foreign travel, if the diarrhoea has not improved by day 7 or the diagnosis is uncertain. Plasma electrolytes, urea, creatinine and glucose should be checked if intravenous fluids are required or there are features suggestive of hypernatraemia. If antibiotics are started, a blood culture should be taken.

Management

This is shown in Figure 13.10.

Hypernatraemic dehydration

The management of hypernatraemic dehydration can be particularly difficult. Oral rehydration solution can be used to rehydrate hypernatraemic children with clinical dehydration. If intravenous fluids are required, a rapid reduction in plasma sodium concentration and osmolality will lead to a shift of water into cerebral cells and may result in seizures and cerebral oedema. The reduction in plasma sodium should therefore be slow. The fluid deficit should be replaced over at least 48 h and the plasma sodium measured regularly, aiming to reduce it at <0.5 mmol/L per hour.

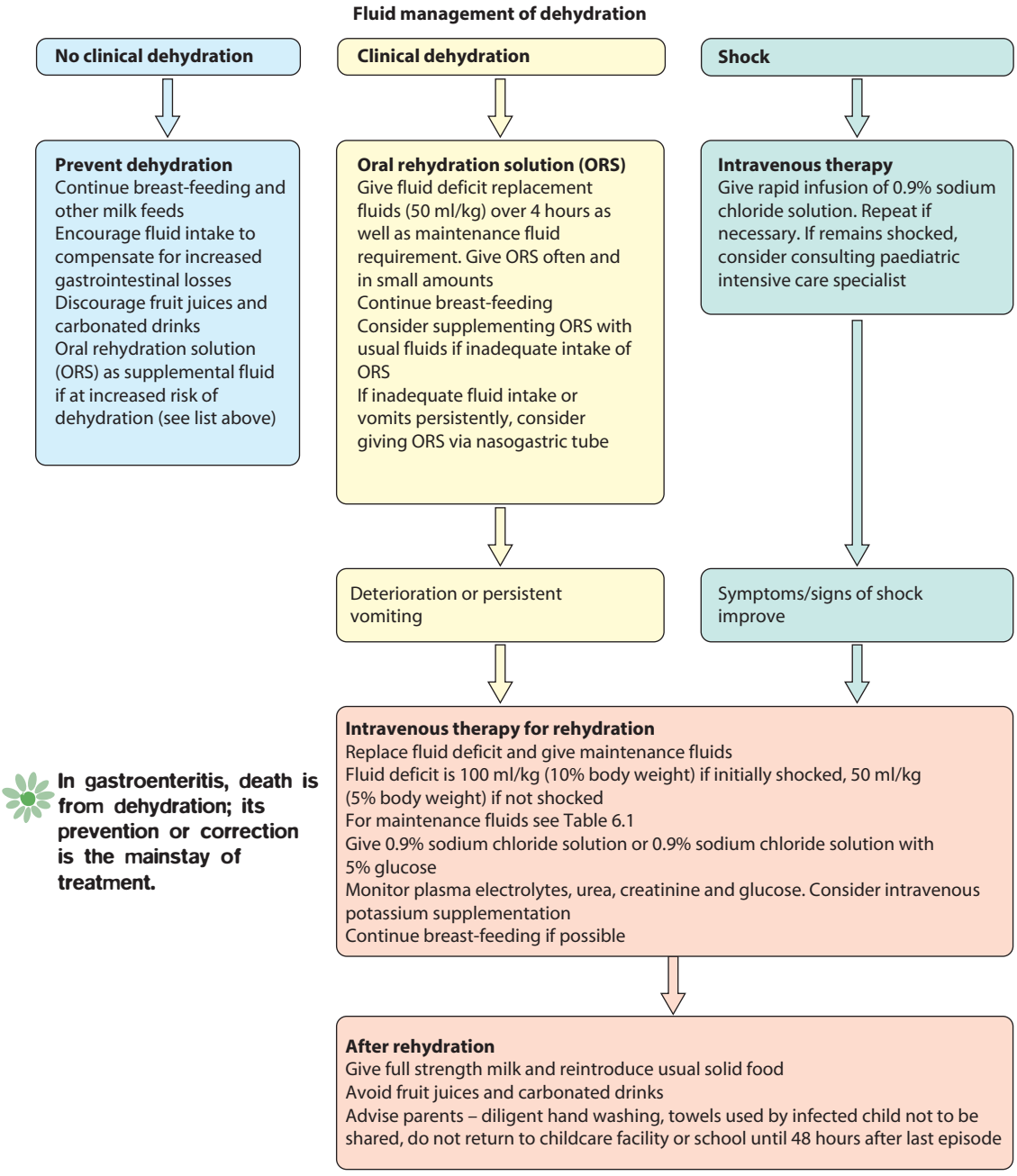
Anti-diarrhoeal drugs (e.g. loperamide, Lomotil) and antiemetics

There is no place for medications for the vomiting or diarrhoea of gastroenteritis as they:

- are ineffective
- may prolong the excretion of bacteria in stools
- can be associated with side-effects
- add unnecessarily to cost
- focus attention away from oral rehydration.

Antibiotics

Antibiotics are not routinely required to treat gastroenteritis, even if there is a bacterial cause. They are only indicated for suspected or confirmed sepsis, extra-intestinal spread of bacterial infection, for salmonella gastroenteritis if <6 months old, in malnourished or immunocompromised children or for specific bacterial or protozoal infections (e.g. *Clostridium difficile*-associated with pseudomembranous colitis, cholera, shigellosis, giardiasis).



In gastroenteritis, death is from dehydration; its prevention or correction is the mainstay of treatment.

Figure 13.10 Fluid management of dehydration. (Adapted from Diarrhoea and vomiting in children, National Institute for Health and Clinical Excellence, 2009).

Nutrition

In developing countries, multiple episodes of diarrhoea are a major contributing factor to the development of malnutrition. Following diarrhoea, nutritional intake should be increased. Diarrhoea may be associated with zinc deficiency and supplementation may be helpful in both acute diarrhoea and as prophylaxis.

Post-gastroenteritis syndrome

Infrequently, following an episode of gastroenteritis, the introduction of a normal diet results in a return of watery diarrhoea. Temporary lactose intolerance may have developed, which can be confirmed by the presence of non-absorbed sugar in the stools giving a positive 'Clinitest' result. In such circumstances, a return to an oral rehydration solution for 24 h, followed by a further introduction of a normal diet, is usually successful.

Rarely, multiple dietary intolerances may result, such that specialist dietary management is required in the implementation of a diet which excludes cow's milk, disaccharides and gluten. In very severe cases, a period of parenteral nutrition is required to enable the injured small intestinal mucosa to recover sufficiently to absorb luminal nutrients.

Summary

Gastroenteritis

Gastroenteritis in developing countries:

- results in death from dehydration of hundreds of thousands of children worldwide every year
- is mostly bacterial from contaminated drinking water and food
- oral rehydration solution – saves the lives of millions of children worldwide each year.

Gastroenteritis in developed countries:

- is mostly viral, but it can be caused by *Campylobacter*, *Shigella* and *Salmonella*
- infants are particularly susceptible to dehydration
- dehydration is assessed as no clinical dehydration, clinical dehydration or shock according to symptoms and signs, but clinical assessment of severity is problematic
- oral rehydration solution is effective in most, but intravenous fluid is required for shock, ongoing vomiting or clinical deterioration.

Malabsorption

Disorders affecting the digestion or absorption of nutrients manifest as:

- abnormal stools
- failure to thrive or poor growth in most but not all cases
- specific nutrient deficiencies, either singly or in combination.

In general, parents know when their children's stools have become abnormal. The true malabsorption stool is difficult to flush down the toilet and has an odour which pervades the whole house. In general, colour is a poor guide to abnormality. Reliable dietetic assessment is important. It is inappropriate to investigate children for malabsorption as a cause of their failure to thrive when dietary energy intake is demonstrably low and other symptoms are absent. Some disorders affecting the small intestinal mucosa or pancreas may lead to the malabsorption of many nutrients (pan-malabsorption), whereas others are highly specific, e.g. zinc malabsorption in *acrodermatitis enteropathica*.

Coeliac disease

Coeliac disease is an enteropathy in which the gliadin fraction of gluten provokes a damaging immunological response in the proximal small intestinal mucosa. As a result, the rate of migration of absorptive cells moving up the villi (enterocytes) from the crypts is massively increased but is insufficient to compensate for increased cell loss from the villous tips. Villi become progressively shorter and then absent, leaving a flat mucosa.

The incidence of 'classical' coeliac disease, diagnosed in childhood on the basis of characteristic clinical symptoms, has been about 1 in 3000 in Europe, including the UK. The age at presentation is partly influenced by the age of introduction of gluten into the diet.

The classical presentation is of a profound malabsorptive syndrome at 8–24 months of age after the introduction of wheat-containing weaning foods. There is failure to thrive, abdominal distension and buttock wasting abnormal stools and general irritability (see Case History 13.2). However, this is no longer the most common presentation and children are now more likely to present less acutely in later childhood. The clinical features of coeliac disease can be highly variable and include mild, non-specific gastrointestinal symptoms, anaemia (iron and/or folate deficiency) and growth failure. Alternatively, it is identified on screening of children at increased risk (type 1 diabetes mellitus, autoimmune thyroid disease, Down syndrome) and first-degree relatives of individuals with known coeliac disease.

The introduction of highly sensitive and specific serological screening tests (IgA tissue transglutaminase antibodies and endomysial antibodies) has provided evidence that coeliac disease is much more common than previously thought and as many as 1 in 100 UK school-age children may be antibody positive.

Diagnosis

Although the diagnosis is strongly suggested by positive serology, confirmation depends upon the demonstration of mucosal changes (increased intraepithelial lymphocytes and a variable degree of villous atrophy and crypt hypertrophy) on small intestinal biopsy performed endoscopically followed by the resolution of symptoms and catch-up growth upon gluten withdrawal.

There is no place for the empirical use of a gluten-free diet as a diagnostic test for coeliac disease in the absence of a small intestinal biopsy. Serological tests such as IgA tissue transglutaminase antibodies and endomysial antibodies are not currently considered sufficiently sensitive and specific to replace biopsy, particularly as the diet is lifelong.

Management

All products containing wheat, rye and barley are removed from the diet and this results in resolution of symptoms. Supervision by a dietician is essential. In children in whom the initial biopsy or the response to gluten withdrawal is doubtful, or when the disease presents before the age of 2 years, a gluten challenge is required in later childhood to demonstrate continuing susceptibility of the small intestinal mucosa to damage by gluten. This is done by either giving food containing gluten or gluten powder. At the start of the challenge, the serological markers should be negative. The gluten challenge is positive if the serology becomes positive again. Repeat small intestinal biopsy is not required. The gluten-free diet should be adhered to for

Case History

13.2 'Classical' coeliac disease

This 2-year-old boy (Fig. 13.11a) had a history of poor growth from 12 months of age (Fig. 13.11b). His parents had noticed that he tended to be crotchety and had three or four foul-smelling stools a day. A jejunal biopsy at 2 years of age showed subtotal villous atrophy (Fig. 13.11c,d) and he was started on a gluten-free diet. Within a few days, his parents commented that his mood had improved and within a month he was a 'different child'. He subsequently exhibited good catch-up growth.



Figure 13.11a Coeliac disease causing wasting of the buttocks and distended abdomen.

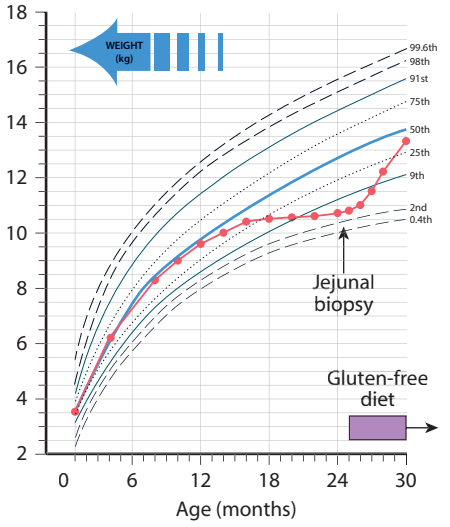


Figure 13.11b Growth chart showing failure to thrive and response to a gluten-free diet. (Adapted from Chart© RCPCH/WHO/Department of Health.)

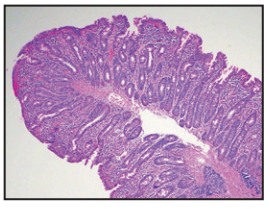


Figure 13.11c Histology of a jejunal biopsy showing lymphocytic infiltration and villous atrophy confirming coeliac disease. (Courtesy of Dr Marie-Anne Brundler.)

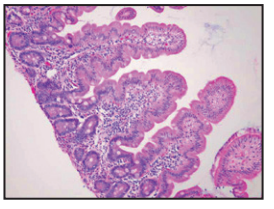


Figure 13.11d Normal jejunal histology is shown for comparison. (Courtesy of Dr Marie-Anne Brundler.)

Summary

Coeliac disease

- A gluten-sensitive enteropathy
- Classical presentation is at 8–24 months with abnormal stools, failure to thrive, abdominal distension, muscle wasting and irritability
- Other modes of presentation – short stature, anaemia, screening, e.g. children with diabetes mellitus
- Diagnosis – positive serology (IgA tissue transglutaminase and endomysial antibodies), flat mucosa on jejunal biopsy and resolution of symptoms and catch-up growth upon gluten withdrawal
- Treatment – gluten-free diet for life.

life. The incidence of small bowel malignancy in adulthood is increased in coeliac disease, although a gluten-free diet probably reduces the risk to normal.

Food allergy and intolerance

This is described in Chapter 15. Allergy.

Other causes of nutrient malabsorption

These are summarised in Figure 13.12.

Toddler diarrhoea

This condition, also called chronic non-specific diarrhoea, is the commonest cause of persistent loose stools in preschool children. Characteristically, the

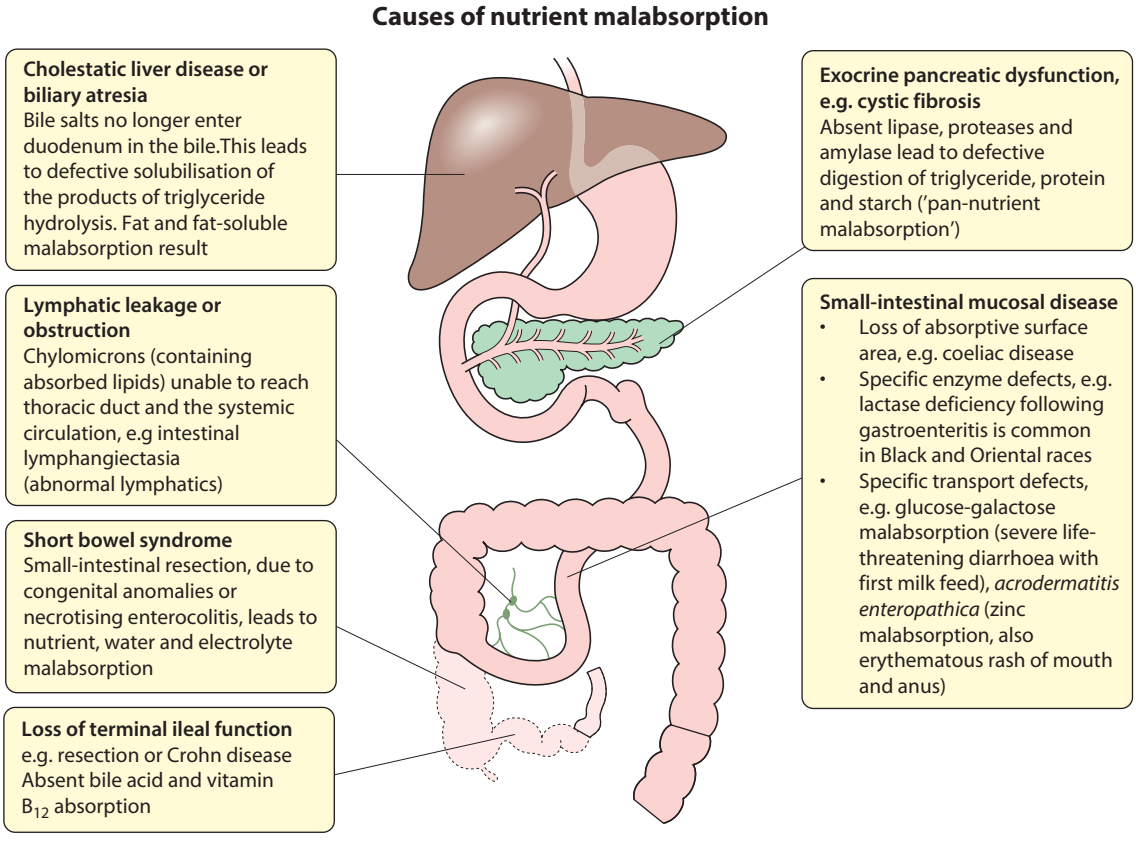


Figure 13.12 Causes of nutrient malabsorption. They are uncommon.

stools are of varying consistency, sometimes well formed, sometimes explosive and loose. The presence of undigested vegetables in the stools is common, giving rise to the alternative title 'peas and carrots diarrhoea'. Affected children are well and thriving and there are no precipitating dietary factors.

Toddler diarrhoea probably results from an underlying maturational delay in intestinal motility which leads to intestinal hurry. The loose stools are not due to malabsorption. Most children have grown out of their symptoms by 5 years of age but achieving faecal continence may be significantly delayed. Some relief of symptoms can be achieved by ensuring that the child's diet contains adequate fat (which slows gut transit) and fibre. Excessive consumption of fresh fruit juice, particularly those high in non-absorbable sorbitol, can exacerbate symptoms.

Summary

Chronic diarrhoea

- In an infant with failure to thrive, consider coeliac disease and cow's milk protein allergy
- Following gastroenteritis, consider post-gastroenteritis syndrome and associated temporary lactose intolerance
- Following bowel resection, cholestatic liver disease or exocrine pancreatic dysfunction, consider malabsorption
- In an otherwise well toddler with undigested vegetables in the stool, consider toddler diarrhoea.

Inflammatory bowel disease

The incidence of inflammatory bowel disease (IBD) in children has increased markedly in the last two decades. The reason for this is unclear. Approximately a quarter of patients present in childhood or adolescence. Crohn disease can affect any part of the gastrointestinal tract from mouth to anus, whereas in ulcerative colitis the inflammation is confined to the colon. Inflammatory bowel disease results from environmental triggers in a genetically predisposed individual. It may cause poor general health, restrict growth and have an adverse effect on psychological well-being. Management requires a specialist multidisciplinary team.

Crohn disease

The clinical features of Crohn disease are summarised in Figure 13.13. Lethargy and general ill health without gastrointestinal symptoms can be the presenting features, particularly in older children. There may be considerable delay in diagnosis as it may be mistaken for